Lecture notes
Advanced Mathematics for Physics

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September 29, 2025

Contents

Chapter 1

General Group Theory

1.1 Group and Subgroup

Definition 1.1.1: Group

A group is a set G equipped with a binary operation

$$\begin{aligned} G \times G &\to G \\ (g, g') &\mapsto gg' \end{aligned}$$

such that:

- **Associativity**: $\forall\, g_1, g_2, g_3 \in G$,

$$(g_1 g_2) g_3 = g_1 (g_2 g_3)$$

- **Identity**: $\exists\, e \in G$ such that $\forall\, g \in G$,

$$eg = ge = g$$

- **Inverses**: $\forall\, g \in G$, $\exists\, g^{-1} \in G$ such that

$$gg^{-1} = g^{-1} g = e$$

Definition 1.1.2: Abelian Group

A group G is *abelian* (or commutative) if and only if $\forall\ g, h \in G$,

$$gh = hg.$$

Definition 1.1.3: Order of a group

A group G is said to be *finite* if and only if G is finite as a set, and its order, denoted $|G|$, is equal to its cardinality $\mathrm{card}(G)$, i.e., the number of elements in G.

Definition 1.1.4: Subgroup

A subgroup H of a group G is a subset $H \subseteq G$ such that:

- **Closure**: $\forall\, h, h' \in H$, $hh' \in H$.
- **Identity**: The identity element e of G is in H, i.e., $e \in H$.
- **Inverses**: $\forall\, h \in H$, $h^{-1} \in H$.

Proposition 1.1.5

A subgroup H of a group G is itself a group

Definition 1.1.6

Let G be a group and let $\Sigma \subseteq G$ be a subset. The *subgroup of* G *generated by* Σ, denoted $\langle\Sigma\rangle$, is the smallest subgroup of G containing Σ. That is, $\langle\Sigma\rangle$ is a subgroup of G such that:

- $\Sigma \subseteq \langle\Sigma\rangle$,
- If H is any subgroup of G with $\Sigma \subseteq H$, then $\langle\Sigma\rangle \subseteq H$.

Example.

Let G be a group and let $g \in G$. The set

$$g^{\mathbb{Z}} = \{g^p : p \in \mathbb{Z}\}$$

is the subgroup of G generated by $\{g\}$, denoted $\langle g\rangle$. To verify this, note that $\langle g\rangle$ is the smallest subgroup of G containing g, and we show:

- $g \in g^{\mathbb{Z}}$, since $g = g^1$ and $1 \in \mathbb{Z}$.
- $g^{\mathbb{Z}}$ is a subgroup:
  - **Closure**: For $g^p, g^q \in g^{\mathbb{Z}}$ with $p, q \in \mathbb{Z}$, $g^p \cdot g^q = g^{p+q}$, and $p + q \in \mathbb{Z}$, so $g^{p+q} \in g^{\mathbb{Z}}$.
  - **Identity**: $e = g^0$ (since $0 \in \mathbb{Z}$), so $e \in g^{\mathbb{Z}}$.
  - **Inverses**: For $g^p \in g^{\mathbb{Z}}$, the inverse is $(g^p)^{-1} = g^{-p}$, and $-p \in \mathbb{Z}$, so $g^{-p} \in g^{\mathbb{Z}}$.
- $g^{\mathbb{Z}}$ is the smallest subgroup containing g: Any subgroup $H \subseteq G$ with $g \in H$ must contain all powers g^p for $p \in \mathbb{Z}$ (by closure and inverses), hence $g^{\mathbb{Z}} \subseteq H$.

Thus, $g^{\mathbb{Z}} = \langle g\rangle$, the cyclic subgroup generated by g.

Definition 1.1.7: Cyclic group

A group G is *cyclic* if and only if it is generated by $\{g\}$ for some $g \in G$.

Proposition 1.1.8

Every cyclic group is abelian.

Proof. Let G be a cyclic group, so $G = \langle g \rangle$ for some $g \in G$. Any element of G is of the form g^p for some $p \in \mathbb{Z}$. Take two elements $g^p, g^q \in G$, where $p, q \in \mathbb{Z}$. Then,

$$g^p \cdot g^q = g^{p+q},$$

since the group operation is exponentiation. Similarly,

$$g^q \cdot g^p = g^{q+p},$$

and since addition in $\mathbb{Z}$ is commutative ($p+q = q+p$), we have $g^{p+q} = g^{q+p}$. Thus, $g^p \cdot g^q = g^q \cdot g^p$, so G is abelian. $\square$

Definition 1.1.9: Cosets and set of cosets

Let G be a group and let $H \subseteq G$ be a subgroup.

- The *left coset* of H in G with respect to an element $g \in G$ is the set

$$gH = \{gh : h \in H\}.$$

  The set of all left cosets of H in G, denoted G/H, is

$$G/H = \{gH : g \in G\}.$$

- The *right coset* of H in G with respect to an element $g \in G$ is the set

$$Hg = \{hg : h \in H\}.$$

  The set of all right cosets of H in G, denoted $H\backslash G$, is

$$H\backslash G = \{Hg : g \in G\}.$$

Remark.

> If H is a subgroup of G that is not normal, then neither G/H nor $H\backslash G$ admits a natural group structure.

1.2 Group homomorphism, conjugation, and normal subgroup

Definition 1.2.1: Group homomorphism

Let G and H be two groups. A *group homomorphism* between G and H is any map

$$f : G \to H$$

such that

$$f(g_1 g_2) = f(g_1)f(g_2)$$

for all $g_1, g_2 \in G$.

Definition 1.2.2: Group isomorphism

Let G and H be groups. A map

$$f : G \to H$$

is a *group isomorphism* if and only if f is a group homomorphism and a bijection. In this case, its inverse $f^{-1} : H \to G$ is also a group isomorphism.

Remark.

> *Notation*: $f : G \xrightarrow{\sim} H$ or $G \cong H$

Definition 1.2.3: Group Automorphism

Let G be a group. A group isomorphism

$$f : G \xrightarrow{\sim} G$$

is called a *group automorphism* of G.

Definition 1.2.4: Automorphism group

The set of all automorphisms of G, denoted $\mathrm{Aut}(G)$, forms a group under function composition, called the *automorphism group of G*.

Definition 1.2.5: Conjugation

Let G be a group. $\forall\, g \in G$, we define the map $\mathcal{C}_g \in \mathrm{Aut}(G)$ such that

$$\mathcal{C}_g(h) = ghg^{-1} \in G \quad \forall\, h \in G.$$

This map is called the *conjugation* by g.

Proof. To show $\mathcal{C}_g \in \mathrm{Aut}(G)$, we must prove it is a group homomorphism and a bijection.

1. **Homomorphism:** We need $\mathcal{C}_g(h_1h_2) = \mathcal{C}_g(h_1)\mathcal{C}_g(h_2)$ for all $h_1, h_2 \in G$. Compute:

$$\mathcal{C}_g(h_1h_2) = g(h_1h_2)g^{-1} = (gh_1g^{-1})(gh_2g^{-1}),$$

since $g^{-1}g = e$ (the identity) and the operation is associative. Now,

$$\mathcal{C}_g(h_1)\mathcal{C}_g(h_2) = (gh_1g^{-1})(gh_2g^{-1}),$$

which matches the above. Thus, $\mathcal{C}_g$ preserves the group operation.

2. **Bijection:** We show $\mathcal{C}_g$ is injective and surjective.

- **Injective:** Suppose $\mathcal{C}_g(h_1) = \mathcal{C}_g(h_2)$. Then:

$$gh_1g^{-1} = gh_2g^{-1}.$$

  Multiply both sides on the left by g^{-1} and on the right by g (using associativity):

$$g^{-1}(gh_1g^{-1})g = g^{-1}(gh_2g^{-1})g \implies (g^{-1}g)h_1(g^{-1}g) = (g^{-1}g)h_2(g^{-1}g) \implies eh_1e = eh_2e \implies h_1 = h_2.$$

  Thus, $\mathcal{C}_g$ is injective.

- **Surjective:** For any $k \in G$, find $h \in G$ such that $\mathcal{C}_g(h) = k$. Set $h = g^{-1}kg$. Then:

$$\mathcal{C}_g(h) = g(g^{-1}kg)g^{-1} = (gg^{-1})k(gg^{-1}) = eke = k.$$

  Since $h = g^{-1}kg \in G$ (as G is closed), $\mathcal{C}_g$ is surjective.

Since $\mathcal{C}_g$ is a homomorphism and bijective, it is an isomorphism from G to itself, hence an automorphism. □

Proposition 1.2.6

Let G be a group. The map

$$\begin{aligned} \mathcal{C} : G &\to \mathrm{Aut}(G) \\ g &\mapsto \mathcal{C}_g \end{aligned}$$

where $\mathcal{C}_g(h) = ghg^{-1}\ \forall\ h \in G$, is a *group homomorphism.*

Proof. We need to show that $\mathcal{C}(g_1g_2) = \mathcal{C}(g_1) \circ \mathcal{C}(g_2)\ \forall\ g_1, g_2 \in G$.
For any $h \in G$,

$$\mathcal{C}_{g_1g_2}(h) = (g_1g_2)h(g_1g_2)^{-1} = g_1g_2hg_2^{-1}g_1^{-1} = g_1(g_2hg_2^{-1})g_1^{-1} = \mathcal{C}_{g_1}(\mathcal{C}_{g_2}(h)).$$

Therefore, $\mathcal{C}_{g_1g_2} = \mathcal{C}_{g_1} \circ \mathcal{C}_{g_2}$, which means $\mathcal{C}(g_1g_2) = \mathcal{C}(g_1) \circ \mathcal{C}(g_2)$.

Since $\mathcal{C}$ maps G to $\mathrm{Aut}(G)$ and preserves the group operation, $\mathcal{C}$ is a group homomorphism. □

Definition 1.2.7: Conjugacy classes

Let G be a group. Two elements $h, h' \in G$ are said to be *conjugate* if there exists an element $g \in G$ such that

$$h = \mathcal{C}_g(h'),$$

where $\mathcal{C}_g(h') = gh'g^{-1}$. This is denoted by $h \sim h'$.
The corresponding equivalence classes are called *conjugacy classes*. For an element $h \in G$, its conjugacy class is defined as

$$[h] = \{h' \in G \mid h' \sim h\}.$$

Definition 1.2.8: Normal Subgroup

Let G be a group, and let $H \subseteq G$ be a subgroup of G. We say that H is a *normal subgroup* of G if and only if

$$\forall g \in G, \quad \mathcal{C}_g(H) \subseteq H,$$

where $\mathcal{C}_g(H) = \{ghg^{-1} \mid h \in H\}$. In fact, $\mathcal{C}_g(H) = H$ for all $g \in G$.

Remark.

> *Notation:* $H \trianglelefteq G$

Definition 1.2.9: Quotient Group

Let G be a group, and let $H \trianglelefteq G$ be a normal subgroup of G. Then:

- The *quotient group* G/H is defined as the set of left (or right) cosets of H in G:

$$G/H = \{gH \mid g \in G\}.$$

- The quotient group G/H is equipped with a binary operation:

$$G/H \times G/H \to G/H,$$

  defined by:

$$(g_1H, g_2H) \mapsto g_1H \cdot g_2H = g_1g_2H.$$

Proof. Let G be a group, and let $H \trianglelefteq G$ be a normal subgroup of G. We show that the quotient group G/H, equipped with the binary operation

$$(g_1H, g_2H) \mapsto g_1H \cdot g_2H = g_1g_2H,$$

satisfies the group axioms.

1. **Closure:** For any $g_1H, g_2H \in G/H$, the product $g_1H \cdot g_2H = g_1g_2H$ is also a coset in G/H. This follows because $g_1g_2 \in G$, so $g_1g_2H \in G/H$.

2. **Associativity:** For any $g_1H, g_2H, g_3H \in G/H$,

$$(g_1H \cdot g_2H) \cdot g_3H = (g_1g_2H) \cdot g_3H = (g_1g_2)g_3H,$$

and

$$g_1H \cdot (g_2H \cdot g_3H) = g_1H \cdot (g_2g_3H) = g_1(g_2g_3)H.$$

Since G is a group, $(g_1g_2)g_3 = g_1(g_2g_3)$, so associativity holds in G/H.

3. **Identity Element:** The coset $eH = H$ acts as the identity element in G/H. For any $gH \in G/H$,

$$eH \cdot gH = egH = gH,$$

and

$$gH \cdot eH = geH = gH.$$

4. **Inverse Element:** For any $gH \in G/H$, the inverse of gH is $g^{-1}H$. This is because

$$gH \cdot g^{-1}H = gg^{-1}H = eH = H,$$

and

$$g^{-1}H \cdot gH = g^{-1}gH = eH = H.$$

Since G/H satisfies all the group axioms, it is a group. $\square$

1.3 First isomorphism theorem

Theorem 1.3.1: First isomorphism theorem

Let G and H be two groups, and let $f : G \to H$ be a group homomorphism. Then:

- The kernel of f, defined as

$$\operatorname{Ker}(f) = \{g \in G \mid f(g) = e_H\},$$

  is a normal subgroup of G, denoted $\operatorname{Ker}(f) \trianglelefteq G$.

- The image of f, denoted $f(G)$, is a subgroup of H and is isomorphic to the quotient group $G/\operatorname{Ker}(f)$. In the case where f is surjective, we have

$$H = f(G) \cong G/\operatorname{Ker}(f).$$

Proof. Let G and H be two groups, and let $f : G \to H$ be a group homomorphism. We prove the two parts of the theorem:

1. **The kernel Ker(f) is a normal subgroup of G:**

   - **Subgroup:**
     - $\operatorname{Ker}(f)$ is nonempty because $f(e_G) = e_H$, so $e_G \in \operatorname{Ker}(f)$.
     - For any $g_1, g_2 \in \operatorname{Ker}(f)$, we have $f(g_1) = e_H$ and $f(g_2) = e_H$. Then:

$$f(g_1 g_2^{-1}) = f(g_1)f(g_2^{-1}) = f(g_1)f(g_2)^{-1} = e_H e_H^{-1} = e_H.$$

     Thus, $g_1 g_2^{-1} \in \operatorname{Ker}(f)$, and $\operatorname{Ker}(f)$ is a subgroup of G.

   - **Normal:** For any $g \in G$ and $k \in \operatorname{Ker}(f)$, we have:

$$f(gkg^{-1}) = f(g)f(k)f(g^{-1}) = f(g)e_H f(g)^{-1} = f(g)f(g)^{-1} = e_H.$$

     Thus, $gkg^{-1} \in \operatorname{Ker}(f)$, and $\operatorname{Ker}(f)$ is normal in G.

2. **The image $f(G)$ is isomorphic to $G/$Ker(f):** Define a map:

$$\phi : G/\operatorname{Ker}(f) \to f(G), \quad \phi(g\operatorname{Ker}(f)) = f(g).$$

   We show that ϕ is an isomorphism:

   - **Well-defined:** If $g_1\operatorname{Ker}(f) = g_2\operatorname{Ker}(f)$, then $g_1^{-1}g_2 \in \operatorname{Ker}(f)$. Thus:

$$f(g_1^{-1}g_2) = e_H \implies f(g_1)^{-1}f(g_2) = e_H \implies f(g_1) = f(g_2).$$

     Therefore, ϕ is well-defined.

   - **Homomorphism:** For any $g_1\operatorname{Ker}(f), g_2\operatorname{Ker}(f) \in G/\operatorname{Ker}(f)$:

$$\phi(g_1\operatorname{Ker}(f) \cdot g_2\operatorname{Ker}(f)) = \phi(g_1 g_2\operatorname{Ker}(f)) = f(g_1 g_2) = f(g_1)f(g_2) = \phi(g_1\operatorname{Ker}(f))\phi(g_2\operatorname{Ker}(f)).$$

Thus, ϕ is a homomorphism.

- **Injective:** If $\phi(g\text{Ker}(f)) = e_H$, then $f(g) = e_H$, so $g \in \text{Ker}(f)$. Thus, $g\text{Ker}(f) = \text{Ker}(f)$, the identity coset. Therefore, ϕ is injective.
- **Surjective:** For any $h \in f(G)$, there exists $g \in G$ such that $f(g) = h$. Then:

$$\phi(g\text{Ker}(f)) = f(g) = h.$$

Thus, ϕ is surjective.

Since ϕ is a bijective homomorphism, it is an isomorphism, and:

$$f(G) \cong G/\text{Ker}(f).$$

If f is surjective, then $f(G) = H$, and:

$$H \cong G/\text{Ker}(f).$$

□

Example.

Let G be a group, and consider the conjugation map:

$$\mathcal{C} : G \to \text{Aut}(G), \quad \mathcal{C}(g) = \mathcal{C}_g,$$

where $\mathcal{C}_g(h) = ghg^{-1}$ for all $h \in G$. Then:

1. The kernel of $\mathcal{C}$ is the center of G:

$$\text{Ker}(\mathcal{C}) = \{g \in G \mid \mathcal{C}_g = \text{id}_G\} = \{g \in G \mid \forall h \in G,\, gh = hg\} = Z(G).$$

2. By the First Isomorphism Theorem, $\text{Ker}(\mathcal{C}) = Z(G)$ is a normal subgroup of G, and:

$$\mathcal{C}(G) \cong G/Z(G).$$

Moreover, $\mathcal{C}(G)$ is a subgroup of $\text{Aut}(G)$.

Proposition 1.3.2

Let G be a group. The conjugation map:

$$\mathcal{C} : G \to \mathrm{Aut}(G), \quad \mathcal{C}(g) = \mathcal{C}_g,$$

where $\mathcal{C}_g(h) = ghg^{-1}$ for all $h \in G$, induces an isomorphism:

$$\mathcal{C}(G) \cong G/Z(G).$$

The group $\mathcal{C}(G)$ is called the **group of inner automorphisms of** G, denoted $\mathrm{Inn}(G)$. Thus:

$$\mathrm{Inn}(G) \cong G/Z(G).$$

The group $\mathrm{Inn}(G)$ is a normal subgroup of $\mathrm{Aut}(G)$, denoted:

$$\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G).$$

The **group of outer automorphisms of** G is defined as the quotient group:

$$\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G).$$

1.4 Direct and Semi-direct products of groups

Definition 1.4.1: Direct products

Let G and H be two groups. The set $G \times H$, equipped with the binary operation:

$$\begin{aligned}(G \times H) \times (G \times H) &\to G \times H \\ (g_1, h_1) \cdot (g_2, h_2) &\mapsto (g_1 g_2, h_1 h_2)\end{aligned}$$

is a group called the **direct product group of** G **and** H, denoted $G \times H$.
The group $G \times H$ admits:

- A normal subgroup isomorphic to G:

$$G \times \{e_H\} = \{(g, e_H) \mid g \in G\} \cong G.$$

- A normal subgroup isomorphic to H:

$$\{e_G\} \times H = \{(e_G, h) \mid h \in H\} \cong H.$$

Definition 1.4.2: Semi-direct products

Let G and H be two groups. Suppose H acts on G by automorphisms, i.e., there exists a group homomorphism:

$$\begin{aligned} f : H &\to \mathrm{Aut}(G) \\ h &\mapsto f_h \end{aligned}$$

where $f_h : G \to G$ is an automorphism for each $h \in H$. The set $G \times H$, equipped with the binary operation:

$$\begin{aligned} (G \times H) \times (G \times H) &\to G \times H \\ (g_1, h_1) \cdot (g_2, h_2) &\mapsto (g_1 f_{h_1}(g_2), h_1 h_2) \end{aligned}$$

is a group called the **semi-direct product of G with H**, denoted $G \rtimes H$.
The group $G \rtimes H$ admits:

- A normal subgroup isomorphic to G:

$$G \times \{e_H\} = \{(g, e_H) \mid g \in G\} \cong G.$$

- A subgroup isomorphic to H:

$$\{e_G\} \times H = \{(e_G, h) \mid h \in H\} \cong H.$$

Example.

> The Euclidean group of affine n-space, denoted $\mathrm{IO}(n)$, is the semi-direct product of the translation group $\mathbb{R}^n$ and the orthogonal group $\mathrm{O}(n)$:
>
> $$\mathrm{IO}(n) = \mathbb{R}^n \rtimes \mathrm{O}(n).$$
>
> Similarly, the special Euclidean group, denoted $\mathrm{ISO}(n)$, is the semi-direct product of $\mathbb{R}^n$ and the special orthogonal group $\mathrm{SO}(n)$:
>
> $$\mathrm{ISO}(n) = \mathbb{R}^n \rtimes \mathrm{SO}(n).$$

Example.

> The Poincaré group, denoted $\mathrm{IO}(1, 3)$, is the semi-direct product of the translation group $\mathbb{R}^4$ and the Lorentz group $\mathrm{O}(1, 3)$:
>
> $$\mathrm{IO}(1, 3) = \mathbb{R}^4 \rtimes \mathrm{O}(1, 3).$$

1.5 Topological groups

1.5.1 A digest of topology

Definition 1.5.1: Topological space

A **topological space** is a pair (X, τ), where X is a set and τ is a set of subsets of X called **open sets**, such that:

- The empty set and the whole set are open:

$$\emptyset, X \in \tau.$$

- τ is stable under finite intersections:

$$\forall \Omega_1, \ldots, \Omega_n \in \tau, \quad \bigcap_{p=1}^{n} \Omega_p \in \tau.$$

- τ is stable under arbitrary unions (finite, countable, or uncountable):

$$\text{For any collection } \{\Omega_\alpha \mid \alpha \in A\} \text{ of open sets in } \tau, \quad \bigcup_{\alpha \in A} \Omega_\alpha \in \tau.$$

Definition 1.5.2: Closed set

Let (X, τ) be a topological space. A subset $U \subseteq X$ is called **closed** if and only if there exists an open set $\Omega \in \tau$ such that:

$$U = X - \Omega.$$

In other words, U is the complement of an open set Ω in X.

Definition 1.5.3: Open cover

Let (X, τ) be a topological space. An **open cover** of X is any collection $\{\Omega_\alpha \mid \alpha \in A\}$ of open sets in τ such that:

$$X = \bigcup_{\alpha \in A} \Omega_\alpha.$$

In other words, the union of all the open sets Ω_α in the collection equals X.

Definition 1.5.4: Husdorff

A topological space (X, τ) is called **Hausdorff** (or **separable**) if and only if for every pair of distinct points $x, y \in X$, there exist open sets $\Omega_x, \Omega_y \in \tau$ such that:

$$x \in \Omega_x, \quad y \in \Omega_y, \quad \text{and} \quad \Omega_x \cap \Omega_y = \emptyset.$$

In other words, any two distinct points in X can be separated by disjoint open sets.

Definition 1.5.5: Compactness

A topological space (X, τ) is called **compact** if and only if every open cover of X has a finite subcover. In other words, for any collection $\{\Omega_\alpha \mid \alpha \in A\}$ of open sets in τ such that:

$$X = \bigcup_{\alpha \in A} \Omega_\alpha,$$

there exists a finite subset $\{\Omega_{\alpha_1}, \dots, \Omega_{\alpha_n}\}$ of the collection such that:

$$X = \bigcup_{i=1}^{n} \Omega_{\alpha_i}.$$

Example.

Let X be a set. The following are examples of topological spaces on X:

- **Trivial Topological Space**: The pair $(X, \{\emptyset, X\})$ is called the **trivial topological space** on X. It consists of only the empty set and the whole set X.
- **Discrete Topological Space**: The pair $(X, 2^X)$, where 2^X is the power set of X (the set of all subsets of X), is called the **discrete topological space** on X. In this topology, every subset of X is open, and in particular, every singleton $\{x\}$ is open for all $x \in X$.
- **Usual Topology on** $\mathbb{R}^n$: Let $n \in \mathbb{N}^*$. The set $\mathbb{R}^n$ equipped with its **usual topology** is a topological space, where a subset $\Omega \subseteq \mathbb{R}^n$ is open if and only if it is either empty or for every $x \in \Omega$, there exists a radius $r > 0$ such that the open ball:

$$B(x, r) = \{y \in \mathbb{R}^n \mid \|y - x\| < r\} \subseteq \Omega.$$

Here, $\|\cdot\|$ denotes the Euclidean norm on $\mathbb{R}^n$.

Theorem 1.5.6: Heine - Borel

A subset $K \subseteq \mathbb{R}^n$ is **compact** (with respect to the usual topology of $\mathbb{R}^n$) if and only if it is **closed** and **bounded**.

Example.

> The space $\mathbb{R}^n$ (with the usual topology) is **non-compact**. This can be seen by considering the open cover:
>
> $$\{B(0,r) \mid r \in \mathbb{R}_+^*\},$$
>
> where $B(0,r) = \{x \in \mathbb{R}^n \mid \|x\| < r\}$ is the open ball of radius r centered at the origin. This open cover has no finite subcover because:
>
> $$\mathbb{R}^n = \bigcup_{r>0} B(0,r),$$
>
> and no finite union of such balls can cover all of $\mathbb{R}^n$.

Example.

> Let (X, τ) be a discrete topological space, where every subset of X is open. Then:
>
> - If X is **finite**, it is compact. This is because any open cover of X has a finite subcover (since X itself is finite).
> - If X is **infinite**, it is not compact. To see this, consider the open cover:
>
> $$\{\{x\} \mid x \in X\}.$$
>
>   This cover has no finite subcover because X is infinite, and each singleton $\{x\}$ contains only one point.

Definition 1.5.7: Continuous Function

Let (X, τ_X) and (Y, τ_Y) be two topological spaces. A function $f : X \to Y$ is called **continuous** if and only if for every open set $\Omega \in \tau_Y$, the preimage:

$$f^{-1}(\Omega) = \{x \in X \mid f(x) \in \Omega\}$$

is an open set in τ_X. In other words:

$$\forall \Omega \in \tau_Y, \quad f^{-1}(\Omega) \in \tau_X.$$

Definition 1.5.8: Connectedness and Simple-Connectedness

Let (X, τ) be a topological space.

- (X, τ) is called **(path)-connected** if and only if for every pair of points $x, y \in X$, there exists a continuous map $\gamma : [0,1] \to X$ such that:

$$\gamma(0) = x \quad \text{and} \quad \gamma(1) = y.$$

  Such a map γ is called a **path** from x to y.

- If (X, τ) is connected, it is called **simply-connected** if and only if every continuous loop $\gamma : [0,1] \to X$ with $\gamma(0) = \gamma(1)$ can be continuously contracted to a point. Otherwise, (X, τ) is called **multiply-connected**.

1.5.2 Topological group

Definition 1.5.9: Topological group

A **topological group** is a tuple $(G, \tau, \cdot)$ such that:

- (G, τ) is a Hausdorff topological space.
- $(G, \cdot)$ is a group.
- The map $g \mapsto g^{-1}$ (inversion) is continuous in the topology τ of G.
- The map $(h, g) \mapsto h \cdot g$ (group multiplication) is continuous in the topology τ of G.

Proposition 1.5.10

Let G be a discrete topological group. Then G is compact if and only if G is finite.

Remark.

> Finite group is just a particular case of compact topological group

Proposition 1.5.11

For every $n \in \mathbb{N}^*$, the general linear group $\mathrm{GL}(n, K)$ is a topological group, where K is either $\mathbb{R}$ or $\mathbb{C}$.

Proof. Let K be either $\mathbb{R}$ or $\mathbb{C}$, and let $\mathrm{GL}(n, K)$ denote the general linear group of $n \times n$ invertible matrices with entries in K. We show that $\mathrm{GL}(n, K)$ is a topological group.

1. **Topological Space**: The group $\mathrm{GL}(n, K)$ is a subspace of K^{n^2}, the space of all $n \times n$ matrices. Since K^{n^2} is Hausdorff (as K is Hausdorff), $\mathrm{GL}(n, K)$ is also Hausdorff.

2. **Group Structure**: The set $\mathrm{GL}(n, K)$ is a group under matrix multiplication, with the identity matrix I_n as the identity element.

3. **Continuity of Multiplication**: The multiplication map $(A, B) \mapsto AB$ is continuous because it is a polynomial function in the entries of A and B. Specifically, each entry of AB is a polynomial in the entries of A and B, and polynomials are continuous functions.

4. **Continuity of Inversion**: The inversion map $A \mapsto A^{-1}$ is continuous because it is a rational function in the entries of A. Specifically, each entry of A^{-1} is given by:

$$(A^{-1})_{ij} = \frac{(-1)^{i+j} \det(A_{ji})}{\det(A)},$$

where A_{ji} is the submatrix obtained by deleting the j-th row and i-th column of A. Since $\det(A) \neq 0$ for $A \in \mathrm{GL}(n, K)$, the inversion map is well-defined and continuous.

5. **Conclusion**: Since $\mathrm{GL}(n, K)$ is a Hausdorff topological space, and both multiplication and inversion are continuous, it is a topological group.

□

Remark.

> ### Definition 1.5.12: Frobenius norm
>
> The **Frobenius norm** is a function $\|\cdot\| : \mathrm{Mat}_n(K) \to K$ defined for a matrix $A \in \mathrm{Mat}_n(K)$ as:
>
> $$\|A\| = \sqrt{\mathrm{Tr}(A^T A)},$$
>
> where $\mathrm{Tr}(A^T A)$ is the trace of the matrix $A^T A$.
> In terms of the components of A, the Frobenius norm can be expressed as:
>
> $$\|A\| = \sqrt{\sum_{i,j=1}^{n} |A_{ij}|^2}.$$
>
> This agrees with the Euclidean norm of K^{n^2}, treating A as a vector in K^{n^2}.

Proposition 1.5.13

$\forall\, n \in \mathbb{N}^*$:

- The unitary group $U(n)$ is a compact topological group.
- The orthogonal group $O(n)$ is a compact topological group.

1.5.3 Identity component and component group

Definition 1.5.14: Identity Component

Let G be a topological group. The **identity component** (or **connected component of the identity**) of G, denoted G^0, is defined as:

$$G^0 = \bigcup_{\substack{\Omega \subseteq G \\ \text{connected},\, e_G \in \Omega}} \Omega,$$

where e_G is the identity element of G, and the union is taken over all connected subsets Ω of G that contain e_G.

Definition 1.5.15: Component group

Let G be a topological group.

- The identity component G^0 is a normal subgroup of G, denoted $G^0 \trianglelefteq G$.
- The **component group** of G, denoted $\pi_0(G)$, is defined as the quotient group:

$$\pi_0(G) = G/G^0.$$

If G^0 is open in G, then $\pi_0(G)$ is a discrete topological group.

*Let G be a topological group, and let G^0 be its identity component. The **component group** $\pi_0(G) = G/G^0$ measures the "disconnectedness" of G in the following ways:*

- ***Number of Connected Components**: The component group $\pi_0(G)$ captures the number of connected components of G. Specifically:*
  - *If G is connected, then $G^0 = G$, and $\pi_0(G)$ is trivial:*

$$\pi_0(G) = \{e\}.$$

  - *If G is disconnected, then $\pi_0(G)$ is non-trivial, and its size corresponds to the number of connected components of G. For example:*
    - *If $\pi_0(G) \cong \mathbb{Z}/2\mathbb{Z}$, then G has two connected components.*
    - *If $\pi_0(G)$ is infinite, then G has infinitely many connected components.*
- ***Structure of Connected Components**: The group structure of $\pi_0(G)$ provides information about how the connected components of G are "arranged" relative to each other. Specifically:*
  - *Each coset gG^0 in $\pi_0(G)$ corresponds to a connected component of G.*
  - *The group operation in $\pi_0(G)$ reflects how these connected components interact under the group operation of G.*
- ***Discrete Topology**: If G^0 is open in G, then the quotient topology on $\pi_0(G)$ is discrete. This means that each connected component of G is "isolated" from the others, and the topology of $\pi_0(G)$ reflects the purely algebraic structure of the group.*

Example.

Consider the general linear group $\mathrm{GL}(1,\mathbb{R})$, which is the multiplicative group of non-zero real numbers:

$$\mathrm{GL}(1,\mathbb{R}) = \mathbb{R}^\times = \{x \in \mathbb{R} \mid x \neq 0\}.$$

- The identity component $\mathrm{GL}(1,\mathbb{R})^0$ is the set of positive real numbers:

  $$\mathrm{GL}(1,\mathbb{R})^0 = \mathbb{R}^\times_+ = \{x \in \mathbb{R} \mid x > 0\}.$$

  This is because $\mathbb{R}^\times_+$ is connected and contains the identity element 1.

- The component group $\pi_0(\mathrm{GL}(1,\mathbb{R}))$ is:

  $$\pi_0(\mathrm{GL}(1,\mathbb{R})) = \mathrm{GL}(1,\mathbb{R})/\mathrm{GL}(1,\mathbb{R})^0 = \{-1, 1\}.$$

  This corresponds to the two connected components of $\mathrm{GL}(1,\mathbb{R})$:

  - The positive real numbers $\mathbb{R}^\times_+$.
  - The negative real numbers $\mathbb{R}^\times_-$.

Example.

The Lorentz group $O(1,3)$ is the group of linear transformations of $\mathbb{R}^4$ that preserve the Minkowski metric:

$$O(1,3) = \{A \in \mathrm{GL}(4,\mathbb{R}) \mid A^T \eta A = \eta\},$$

where $\eta = \mathrm{diag}(-1,1,1,1)$ is the Minkowski metric.

- The identity component $O(1,3)^0$ is the **orthochronous Lorentz group** $\mathrm{SO}^+(1,3)$, which consists of Lorentz transformations that preserve both the orientation of space and the direction of time.

- The component group $\pi_0(O(1,3))$ is:

  $$\pi_0(O(1,3)) = O(1,3)/O(1,3)^0 \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

  This corresponds to the four connected components of $O(1,3)$, which are determined by:

  - Whether the transformation preserves or reverses the orientation of space.
  - Whether the transformation preserves or reverses the direction of time.

1.5.4 Universal covering group and fundamental group

Definition 1.5.16: Universal covering group

Let G be a connected topological group. The **universal covering group** of G, denoted $\widetilde{G}$, is a simply connected topological group equipped with a continuous surjective group homomorphism:

$$p : \widetilde{G} \to G,$$

such that:

- $\widetilde{G}$ is simply connected.
- p is a continuous surjective group homomorphism.
- The kernel of p is a discrete subgroup of $\widetilde{G}$.

Definition 1.5.17: Fundamental group

Let G be a connected topological group with universal covering group $\widetilde{G}$ and covering homomorphism:

$$p : \widetilde{G} \to G.$$

- The **fundamental group** of G, denoted $\pi_1(G)$, is defined as the kernel of p:

  $$\pi_1(G) = \ker(p) \trianglelefteq \widetilde{G}.$$

  It is a discrete normal subgroup of $\widetilde{G}$.

- The universal covering group $\widetilde{G}$ is isomorphic to the quotient group:

  $$\widetilde{G} \cong G/\pi_1(G).$$

1.5.5 Haar integral

Definition 1.5.18: Haar measure

Let G be a compact topological group, and let $f : G \to K$ be a continuous function, where K is a field (typically $\mathbb{R}$ or $\mathbb{C}$).

- If G is finite, the **average** of f on G, denoted $\langle f \rangle$, is defined as:

$$\langle f \rangle = \frac{1}{|G|} \sum_{g \in G} f(g).$$

- The average is both left and right invariant:
  - **Left invariance**: For any $h \in G$,

$$\langle f(h\cdot) \rangle = \frac{1}{|G|} \sum_{g \in G} f(hg) = \frac{1}{|G|} \sum_{g \in G} f(g) = \langle f \rangle.$$

  - **Right invariance**: For any $h \in G$,

$$\langle f(\cdot h) \rangle = \frac{1}{|G|} \sum_{g \in G} f(gh) = \frac{1}{|G|} \sum_{g \in G} f(g) = \langle f \rangle.$$

Example.

The circle group $U(1) = \{e^{i\theta} \mid \theta \in [0, 2\pi)\}$ is a compact topological group. Its Haar measure is given by the normalized arc length measure on the circle:

$$\langle f \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta})\, d\theta,$$

where $f : U(1) \to \mathbb{C}$ is a continuous function. This measure satisfies:

- **Left/Right Invariance**: For any fixed $\phi \in [0, 2\pi)$, both

$$\frac{1}{2\pi} \int_0^{2\pi} f(e^{i(\theta+\phi)})\, d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta})\, d\theta$$

  and

$$\frac{1}{2\pi} \int_0^{2\pi} f(e^{i(\phi+\theta)})\, d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta})\, d\theta$$

  hold, demonstrating rotation invariance.

- **Normalization**:

$$\frac{1}{2\pi} \int_0^{2\pi} 1\, d\theta = 1,$$

  making it a probability measure.

Theorem 1.5.19: Haar Integral

For any compact topological group G, there exists a unique (up to normalization) integral called the **Haar integral** satisfying:

- **Left and Right Invariance**: For any continuous function $f : G \to \mathbb{C}$ and any $h \in G$,

$$\int_G f(hg)\, dg = \int_G f(gh)\, dg = \int_G f(g)\, dg.$$

- **Normalization**: The integral can be normalized so that

$$\int_G 1\, dg = 1,$$

  making G a probability space.

This integral is the unique translation-invariant measure on G.

Proposition 1.5.20

Let G be a compact topological group with Haar measure dg, and let $f : G \to \mathbb{R}$ be a continuous function. If:

$$\int_G f(g)\, dg = 0,$$

then f is identically zero on G (i.e., $f(g) = 0$ for all $g \in G$).

Remark.

For a Lie group G that is locally diffeomorphic to $\mathbb{R}^n$, the Haar integral can be expressed in local coordinates $(x^1, ..., x^n)$ as:

$$\int_G f(g)\, dg = \int_{\mathbb{R}^n} f(x)\rho(x)\, dx^1 \cdots dx^n,$$

where:

- $\rho(x)$ is a density function accounting for the group structure
- $dx^1 \cdots dx^n$ is the Lebesgue measure on $\mathbb{R}^n$
- The density $\rho(x)$ ensures left/right invariance

Example.

1. For $G = \mathbb{R}^n$ (additive group):

$$\rho(x) \equiv 1 \quad \text{(Lebesgue measure)}$$

2. For $G = \mathrm{GL}(n, \mathbb{R})$:

$$\rho(A) = |\det A|^{-n} \quad \text{(matrix coordinates)}$$

3. For $G = \mathrm{SO}(3)$ (Euler angles α, β, γ):

$$\rho(\alpha, \beta, \gamma) = \sin\beta \quad \text{(standard rotation measure)}$$

Chapter 2

Representation theory of groups

2.1 Representation of topological groups

Definition 2.1.1: Representation

Let G be a group (or topological group) and V a vector space over a field K (typically $K = \mathbb{C}$ or $\mathbb{R}$). A **representation** of G on V is a continuous group homomorphism:

$$\rho : G \to \mathrm{GL}(V),$$

where:

- $\mathrm{GL}(V)$ is the general linear group of invertible linear transformations on V
- For each $g \in G$, $\rho(g)$ is an invertible linear operator on V
- The map ρ preserves the group operation: $\rho(gh) = \rho(g)\rho(h)$

The **dimension** of ρ, denoted $\dim \rho$, is the dimension of V as a K-vector space:

$$\dim \rho := \dim_K V.$$

We say ρ is **finite-dimensional** if $\dim_K V < \infty$.

Remark.

- For topological groups, continuity is with respect to the topology on G and the operator norm topology on $\mathrm{GL}(V)$
- When $\dim V = n$, $\mathrm{GL}(V) \cong \mathrm{GL}(n, K)$
- A representation makes abstract group elements concrete as matrices/operators

Definition 2.1.2: Trivial representation

The **trivial representation** of a group G on a vector space V is the continuous group homomorphism:

$$\begin{aligned} \rho_{\text{triv}} : G &\to \mathrm{GL}(V) \\ g &\mapsto \mathrm{id}_V \end{aligned}$$

where id_V is the identity operator on V.

Remark.

- **Stability**: The trivial representation is *stable* under group actions, meaning:

  $$\rho_{\text{triv}}(g)(v) = v \quad \forall g \in G, \forall v \in V$$

  No non-zero vector is moved by any group element.

- **Irreducibility**: If $\dim V = 1$, the trivial representation is irreducible (no non-trivial invariant subspaces).

- **Universality**: Appears in the decomposition of every representation as a summand when G acts trivially on a subspace.

Definition 2.1.3: ρ-Stable Subspace

Let $\rho : G \to \mathrm{GL}(V)$ be a representation of a group G on a K-vector space V. A subspace $U \subseteq V$ is called **ρ-stable** (or **G-invariant**) if:

$$\forall g \in G, \quad \rho(g)(U) \subseteq U.$$

For such a subspace, the **subrepresentation** $\rho|_U : G \to \mathrm{GL}(U)$ is defined by:

$$\rho|_U(g) := \rho(g)\big|_U,$$

where $\rho(g)\big|_U$ denotes the restriction of $\rho(g)$ to U.

Remark.

- **Trivial Subspaces**:
  - $\{0\}$ and V are always ρ-stable for any representation ρ.
- **Irreducibility**:
  - If V has no non-trivial ρ-stable subspaces (i.e., only $\{0\}$ and V are stable), ρ is called **irreducible**.
- **Constructing New Representations**:
  - If U is ρ-stable, the quotient space V/U inherits a **quotient representation** $\rho_{V/U}$.

Definition 2.1.4: Irreducible Representation

A representation $\rho : G \to \mathrm{GL}(V)$ is called **irreducible** if the only ρ-stable subspaces of V are the trivial ones:

$$\{0\} \quad \text{and} \quad V.$$

In other words, V has no non-trivial proper subspaces preserved by all $\rho(g)$ for $g \in G$.
Equivalent Characterization:

$$\rho \text{ is irreducible} \iff \nexists \text{ proper subspace } U \subsetneq V \text{ with } \rho(g)(U) \subseteq U \text{ for all } g \in G.$$

Remark.

- **Atomic Building Blocks**: Irreducible representations are the “simplest” representations that cannot be decomposed further into smaller stable subspaces.
- **Schur’s Lemma**: If ρ is irreducible, any intertwining operator $T : V \to V$ (i.e., satisfying $T \circ \rho(g) = \rho(g) \circ T$) is a scalar multiple of the identity.
- **Complete Reducibility**: For finite groups (or compact topological groups), every representation decomposes as a direct sum of irreducibles.

Definition 2.1.5: Equivalent representations

Two representations of a group G:

$$\rho_1 : G \to \mathrm{GL}(V_1) \quad \text{and} \quad \rho_2 : G \to \mathrm{GL}(V_2)$$

are called **equivalent** (denoted $\rho_1 \sim \rho_2$) if there exists a linear isomorphism

$$\phi : V_1 \xrightarrow{\sim} V_2$$

such that:

$$\phi \circ \rho_1(g) = \rho_2(g) \circ \phi \quad \forall g \in G.$$

Remark.

- **Intertwining Condition**: The map ϕ is called an **intertwining operator** between ρ_1 and ρ_2.
- **Invariants Preserved**:
  - Dimensions: $\dim V_1 = \dim V_2$
  - Character values (for finite groups): $\chi_{\rho_1}(g) = \chi_{\rho_2}(g)$
  - Irreducibility: ρ_1 is irreducible iff ρ_2 is
- **Notation**: When ϕ is specified, we write $\rho_1 \overset{\phi}{\sim} \rho_2$.

Definition 2.1.6: Unitary Representation

A representation $\rho : G \to \mathrm{GL}(V)$ of a group G on a complex Hilbert space V is called **unitary** if V admits a Hermitian inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$ satisfying:

$$\forall g \in G, \forall x, y \in V, \quad \langle \rho(g)x, \rho(g)y \rangle = \langle x, y \rangle.$$

Equivalently:

- Each operator $\rho(g)$ is unitary: $\rho(g)^* \rho(g) = \mathrm{id}_V$.
- The representation lands in the unitary group: $\rho : G \to \mathrm{U}(V)$, where

$$\mathrm{U}(V) := \{T \in \mathrm{GL}(V) \mid T^* T = \mathrm{id}_V\}.$$

Remark.

- **Preservation of Structure**: Unitary representations preserve quantum probabilities (Born rule):
$$|\langle \rho(g)\psi, \rho(g)\phi\rangle|^2 = |\langle \psi, \phi\rangle|^2.$$
- **Complete Reducibility**: For compact G, every unitary representation decomposes into orthogonal irreducibles:
$$V \cong \bigoplus_k V_k^{\oplus n_k}.$$
- **Adjoint Action**: The adjoint representation satisfies $\rho(g^{-1}) = \rho(g)^*$.

Remark.

1. **Continuity of Finite-Dimensional Representations**:
If $\dim \rho < \infty$, the continuity of $\rho : G \to \mathrm{GL}(V)$ is equivalent to the continuity of all matrix coefficient maps:
$$g \mapsto \langle e_i, \rho(g)e_j\rangle \in C^0(G), \quad \forall i, j$$
where $\{e_i\}$ is a basis for V.
2. **Representations of Discrete Groups**:
If G is discrete, every homomorphism $\rho : G \to \mathrm{GL}(V)$ is automatically continuous, as the discrete topology requires no convergence conditions.
3. **Finite vs. Infinite-Dimensional Representations**:
While infinite-dimensional representations exist, finite-dimensional representations are often studied first due to:
   - Simplicity of operator theory in finite dimensions
   - Applications to quantum mechanics and geometry

Theorem (Corrected): Peter-Weyl Theorem (Simplified)
Let G be a compact topological group. Then:

- Every *irreducible* unitary representation of G is finite-dimensional.
- The regular representation $L^2(G)$ decomposes into an orthogonal direct sum of finite-dimensional irreducible representations.

Counterexample to Original Claim:
The left regular representation $\rho : G \to \mathrm{U}(L^2(G))$, defined by:
$$(\rho(g)f)(h) = f(g^{-1}h),$$
is infinite-dimensional for non-finite compact G.

2.1.1 Contra-gradient representations

Definition 2.1.7: Dual Space

Let V be a vector space over a field $\mathbb{K}$. A *linear form* (or *linear functional*) on V is a linear map

$$\theta\colon V \to \mathbb{K}$$

The set V^* of all linear forms on V forms a $\mathbb{K}$-vector space called the *dual space* of V.

Definition 2.1.8: Contragradient Representation

Let G be a topological group and V a $\mathbb{K}$-vector space. Given a representation

$$\rho\colon G \to \mathrm{GL}(V)$$

the *contragradient representation*

$$\rho^*\colon G \to \mathrm{GL}(V^*)$$

is defined by:

$$\rho^*(g)(\theta)(v) := \theta\left(\rho(g^{-1})(v)\right)$$

for all $g \in G$, $\theta \in V^*$, and $v \in V$.

Remark.

This construction:

- Preserves the group structure (ρ^* is a homomorphism)
- Makes V^* a G-module via the dual action
- Is sometimes called the dual or contravariant representation

Proof. We verify that ρ^* is a well-defined representation:

1. **Linearity**: For all $g \in G$, $\theta \in V^*$, and $v \in V$,

$$\rho^*(g)(\theta)(v) = \theta(\rho(g^{-1})v) \in \mathbb{K}.$$

Since $\rho(g^{-1})$ is linear and θ is linear, $\rho^*(g)(\theta)$ is a linear functional on V, so $\rho^*(g)(\theta) \in V^*$.

2. **Group Homomorphism**: For all $g, h \in G$, $\theta \in V^*$, and $v \in V$,

$$\begin{aligned}
\rho^*(gh)(\theta)(v) &= \theta(\rho((gh)^{-1})v) \\
&= \theta(\rho(h^{-1}g^{-1})v) \\
&= \theta(\rho(h^{-1})\rho(g^{-1})v) \quad \text{(since } \rho \text{ is a homomorphism)} \\
&= \rho^*(h)(\theta)(\rho(g^{-1})v) \quad \text{(by definition of } \rho^*\text{)} \\
&= \rho^*(g)\left(\rho^*(h)(\theta)\right)(v) \\
&= (\rho^*(g) \circ \rho^*(h))(\theta)(v).
\end{aligned}$$

Thus, $\rho^*(gh) = \rho^*(g) \circ \rho^*(h)$, proving ρ^* is a group homomorphism.

3. **Continuity (for topological groups)**: If G is a topological group and ρ is continuous, then ρ^* is continuous because:

- $\rho(g^{-1})$ depends continuously on g,
- The evaluation map $\theta(v)$ is continuous in θ for finite-dimensional V,
- Finite-dimensionality ensures $V \cong V^{**}$, making the dual action continuous.

□

2.1.2 Operations on representations

Definition 2.1.9: Direct sum of representations

Let G be a topological group, and let V_1, V_2 be $\mathbb{K}$-vector spaces. Given two representations:

$$\rho_1 \colon G \to \mathrm{GL}(V_1), \quad \rho_2 \colon G \to \mathrm{GL}(V_2),$$

their **direct sum representation**

$$\rho_1 \oplus \rho_2 \colon G \to \mathrm{GL}(V_1 \oplus V_2)$$

is defined by the action:

$$(\rho_1 \oplus \rho_2)(g)(v_1, v_2) := \big(\rho_1(g)v_1,\, \rho_2(g)v_2\big)$$

for all $g \in G$, $v_1 \in V_1$, and $v_2 \in V_2$.
This construction satisfies:

- $\rho_1 \oplus \rho_2$ is a valid representation of G.
- $\dim(\rho_1 \oplus \rho_2) = \dim(\rho_1) + \dim(\rho_2)$.

Remark.

> Given representations $\rho_1 \colon G \to \mathrm{GL}(V_1)$ and $\rho_2 \colon G \to \mathrm{GL}(V_2)$, in chosen bases for V_1 and V_2, the matrix of their direct sum representation is block diagonal:
>
> $$(\rho_1 \oplus \rho_2)(g) = \begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix}$$

Example.

> For $G = \mathbb{R}$ and:
>
> $$\rho_1(t) = \begin{pmatrix} e^t \end{pmatrix}, \quad \rho_2(t) = \begin{pmatrix} e^{-t} \end{pmatrix},$$
>
> the direct sum acts on $\mathbb{R}^2$ as:
>
> $$(\rho_1 \oplus \rho_2)(t) = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

Definition 2.1.10: Tensor product of representations

Let G be a topological group and V_1, V_2 be $\mathbb{K}$-vector spaces. Given two representations:

$$\rho_1 \colon G \to \mathrm{GL}(V_1), \quad \rho_2 \colon G \to \mathrm{GL}(V_2),$$

their **tensor product representation**

$$\rho_1 \otimes \rho_2 \colon G \to \mathrm{GL}(V_1 \otimes V_2)$$

is defined by the action:

$$(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) := \rho_1(g)v_1 \otimes \rho_2(g)v_2$$

for all $g \in G$, $v_1 \in V_1$, and $v_2 \in V_2$, extended linearly to all of $V_1 \otimes V_2$.
This construction satisfies:

- $\rho_1 \otimes \rho_2$ is a well-defined representation of G
- $\dim(\rho_1 \otimes \rho_2) = \dim(\rho_1) \cdot \dim(\rho_2)$

Proof. **Part 1: Well-definedness of the action**
For each $g \in G$, define the map:

$$\phi_g : V_1 \times V_2 \to V_1 \otimes V_2, \quad (v_1, v_2) \mapsto \rho_1(g)v_1 \otimes \rho_2(g)v_2.$$

This map is:

- **Bilinear**: Since $\rho_1(g)$ and $\rho_2(g)$ are linear, ϕ_g preserves linearity in each argument:

$$\phi_g(\lambda v_1 + v_1', v_2) = \lambda \rho_1(g)v_1 \otimes \rho_2(g)v_2 + \rho_1(g)v_1' \otimes \rho_2(g)v_2 = \lambda \phi_g(v_1, v_2) + \phi_g(v_1', v_2).$$

Similarly for the second argument.

By the universal property of tensor products, ϕ_g induces a unique *linear* map:

$$(\rho_1 \otimes \rho_2)(g) : V_1 \otimes V_2 \to V_1 \otimes V_2.$$

Part 2: Homomorphism property

For $g, h \in G$ and $v_1 \otimes v_2 \in V_1 \otimes V_2$:

$$\begin{aligned}
(\rho_1 \otimes \rho_2)(gh)(v_1 \otimes v_2) &= \rho_1(gh)v_1 \otimes \rho_2(gh)v_2 \\
&= \rho_1(g)\rho_1(h)v_1 \otimes \rho_2(g)\rho_2(h)v_2 \quad \text{(since } \rho_1, \rho_2 \text{ are homomorphisms)} \\
&= (\rho_1 \otimes \rho_2)(g)\,(\rho_1(h)v_1 \otimes \rho_2(h)v_2) \\
&= (\rho_1 \otimes \rho_2)(g) \circ (\rho_1 \otimes \rho_2)(h)(v_1 \otimes v_2).
\end{aligned}$$

Thus, $\rho_1 \otimes \rho_2$ preserves the group operation.

Part 3: Invertibility

For each $g \in G$, the inverse of $(\rho_1 \otimes \rho_2)(g)$ is $(\rho_1 \otimes \rho_2)(g^{-1})$:

$$(\rho_1 \otimes \rho_2)(g) \circ (\rho_1 \otimes \rho_2)(g^{-1}) = (\rho_1 \otimes \rho_2)(gg^{-1}) = (\rho_1 \otimes \rho_2)(e) = \mathrm{Id}_{V_1 \otimes V_2}.$$

Part 4: Dimension

If $\dim(V_1) = n$ and $\dim(V_2) = m$, then $\dim(V_1 \otimes V_2) = n \cdot m$. The matrix of $(\rho_1 \otimes \rho_2)(g)$ in a basis $\{e_i \otimes f_j\}$ is the Kronecker product of the matrices of $\rho_1(g)$ and $\rho_2(g)$, hence has size $nm \times nm$.

$\square$

Remark.

Let G be a topological group. We define:

- $\mathrm{Rep}(G)$ to be the set of isomorphism classes of finite-dimensional continuous linear representations of G over $\mathbb{C}$,
- $\oplus$ as the direct sum operation on representations,
- $\otimes$ as the tensor product operation on representations.

The triple $(\mathrm{Rep}(G), \oplus, \otimes)$ forms a commutative ring called the **representation ring** (or **Green ring**) of G, where:

- Addition $\oplus$ corresponds to direct sum of representations,
- Multiplication $\otimes$ corresponds to tensor product of representations,
- The additive identity is the zero representation,
- The multiplicative identity is the trivial 1-dimensional representation.

The fundamental objective of representation theory is to:

- Completely describe the structure of $\mathrm{Rep}(G)$ as a ring for a given group G,
- Understand the irreducible representations as the "building blocks" of $\mathrm{Rep}(G)$,
- Relate the ring structure to the group structure of G through character theory.

2.2 Representation theory for compact topological group

2.2.1 Complete Reducibility

Definition 2.2.1: Unitary Representation Reducibility

Let:

- G be a topological group (not necessarily compact),
- V be a $\mathbb{K}$-vector space equipped with a Hermitian inner product $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{K}$,
- $\rho \colon G \to \mathrm{U}(V, \langle \cdot, \cdot \rangle)$ be a unitary representation of G on V.

If $U \subseteq V$ is a ρ-invariant subspace (i.e., $\rho(g)U \subseteq U$ for all $g \in G$), then:

1. The orthogonal complement $U^\perp$ is also ρ-invariant,
2. The representation decomposes as:

$$\rho \simeq \rho|_U \oplus \rho|_{U^\perp}$$

where $\rho|_U$ and $\rho|_{U^\perp}$ are the subrepresentations restricted to U and $U^\perp$ respectively.

Proposition 2.2.2

Let G be a compact topological group and V a finite-dimensional $\mathbb{K}$-vector space (where $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$). For any continuous representation $\rho\colon G \to \mathrm{GL}(V)$, there exists a G-invariant Hermitian (resp. positive-definite symmetric if $\mathbb{K} = \mathbb{R}$) inner product $\langle \cdot, \cdot \rangle$ on V that makes ρ unitary (resp. orthogonal). That is:

$$\langle \rho(g)v, \rho(g)w \rangle = \langle v, w \rangle \quad \forall g \in G, \forall v, w \in V.$$

Proof. We construct the invariant inner product using Haar integration:

1. **Existence of Haar Measure**: Since G is compact, it admits a unique bi-invariant Haar measure μ with $\mu(G) = 1$.

2. **Averaging an Arbitrary Inner Product**: Fix any Hermitian inner product $(\cdot, \cdot)$ on V. Define:

$$\langle v, w \rangle := \int_G (\rho(g)v, \rho(g)w)\, d\mu(g).$$

3. **Invariance Verification**: For any $h \in G$:

$$\langle \rho(h)v, \rho(h)w \rangle = \int_G (\rho(g)\rho(h)v, \rho(g)\rho(h)w)\, d\mu(g) = \int_G (\rho(gh)v, \rho(gh)w)\, d\mu(g).$$

By right-invariance of μ, this equals:

$$\int_G (\rho(g)v, \rho(g)w)\, d\mu(g) = \langle v, w \rangle.$$

4. **Positive-Definiteness**: $\langle v, v \rangle \geq 0$ with equality iff $v = 0$, since $(\cdot, \cdot)$ is positive-definite and μ is positive. □

Proposition 2.2.3

Let G be a compact topological group. Every finite-dimensional continuous representation $\rho\colon G \to \mathrm{GL}(V)$ decomposes as a direct sum of irreducible representations. That is, there exists a G-equivariant isomorphism:

$$V \cong V_1 \oplus \cdots \oplus V_n$$

where each (V_i, ρ_i) is an irreducible subrepresentation.

Proof. We proceed in three steps:

1. **Unitarization**: By the previous proposition, there exists a G-invariant Hermitian inner product $\langle \cdot, \cdot \rangle$ on V making ρ unitary.

2. **Inductive Decomposition**:

- If ρ is irreducible, we are done.
- Otherwise, let $W \subset V$ be a nontrivial proper G-invariant subspace.
- By unitarity, $V = W \oplus W^{\perp}$ where $W^{\perp}$ is also G-invariant.

- Apply the same argument recursively to W and $W^{\perp}$.

3. **Termination**: The process terminates in finitely many steps since $\dim V < \infty$. □

Remark.

Let G be a compact topological group. The representation ring $(\mathrm{Rep}(G), \oplus, \otimes)$ is generated as a $\mathbb{Z}$-algebra by its irreducible representations. That is:

- Every finite-dimensional continuous representation ρ decomposes into a finite direct sum of irreducibles
- The tensor product of any two representations can be expressed in terms of irreducible components

To completely determine $\mathrm{Rep}(G)$*, one must:*

1. ***Classify irreducible representations****: Find a complete set* $\{\rho_i\}_{i\in I}$ *of pairwise non-isomorphic irreducible representations.*
2. ***Solve the Clebsch-Gordan problem****: For each pair* (ρ_i, ρ_j)*, determine the Clebsch-Gordan coefficients* $m_{i,j}^k \in \mathbb{N}$ *in the decomposition:*

$$\rho_i \otimes \rho_j \cong \bigoplus_{k\in I} \rho_k^{\oplus m_{i,j}^k}$$

This defines the tensor product table (or fusion rules) of G*.*

Remark.

For arbitrary representations $\rho, \rho' \in \mathrm{Rep}(G)$, we have:

$$\begin{aligned}
\rho &\cong \bigoplus_{i\in I} \rho_i^{\oplus m_i(\rho)} \\
\rho' &\cong \bigoplus_{j\in I} \rho_j^{\oplus m_j(\rho')} \\
\rho \otimes \rho' &\cong \bigoplus_{i,j\in I} (\rho_i \otimes \rho_j)^{\oplus m_i(\rho) m_j(\rho')} \\
&\cong \bigoplus_{k\in I} \rho_k^{\oplus\left(\sum_{i,j} m_i(\rho) m_j(\rho') m_{i,j}^k\right)}
\end{aligned}$$

where:

- $m_i(\rho) = \dim \mathrm{Hom}_G(\rho_i, \rho)$ are multiplicity coefficients
- $m_{i,j}^k$ are the Clebsch-Gordan coefficients from the irreducible tensor products

Example.

Consider the additive group $(\mathbb{R}, +)$ with its standard topology. Define a 2-dimensional representation:

$$\rho\colon \mathbb{R} \to \mathrm{GL}(2, \mathbb{R}), \quad \rho(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Verification of Representation Properties:

1. **Group Homomorphism**: For all $x, y \in \mathbb{R}$,

$$\begin{aligned} \rho(x+y) &= \begin{pmatrix} 1 & x+y \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & y \\ 0 & 1 \end{pmatrix} \\ &= \rho(x)\rho(y) \end{aligned}$$

2. **Continuity**: The map $x \mapsto \rho(x)$ is continuous in the Euclidean topology.

Reducibility Analysis:

- **Reducible**: The subspace $W = \mathbb{R}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is ρ-invariant since:

$$\rho(x)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in W.$$

- **Indecomposable**: Suppose there exists another invariant subspace U complementary to W. Then U must be spanned by a vector $\begin{pmatrix} a \\ b \end{pmatrix}$ with $b \neq 0$. However:

$$\rho(x)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+xb \\ b \end{pmatrix} \notin \operatorname{span}\left\{\begin{pmatrix} a \\ b \end{pmatrix}\right\} \quad \text{for } x \neq 0,$$

unless $b = 0$, which contradicts $U \cap W = \{0\}$. Thus, ρ admits no complementary invariant subspace.

Conclusion:

- ρ is *reducible* (has a nontrivial invariant subspace W).
- ρ is *indecomposable* (cannot be expressed as $W \oplus U$ for another invariant subspace U).
- This contrasts starkly with the behavior of compact group representations, which are always completely reducible.

Characterization: The representation ρ is a *nontrivial extension*:

$$0 \to W \to \mathbb{R}^2 \to \mathbb{R}^2/W \to 0,$$

where both W and $\mathbb{R}^2/W$ are trivial representations. This extension does not split, demonstrating the failure of complete reducibility for non-compact groups.

2.2.2 Schur's lemma and Schur's orthogonality

Lemma 2.2.4: Schur

Let G be a topological group and V_1, V_2 be finite-dimensional $\mathbb{K}$-vector spaces ($\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$). Given two irreducible continuous representations:

$$\rho_1 \colon G \to \mathrm{GL}(V_1), \quad \rho_2 \colon G \to \mathrm{GL}(V_2),$$

and a linear map $\phi \colon V_1 \to V_2$ satisfying the **intertwining property**:

$$\rho_2(g) \circ \phi = \phi \circ \rho_1(g) \quad \text{for all } g \in G,$$

then exactly one of the following holds:

1. ϕ is an isomorphism (and thus $\rho_1 \simeq \rho_2$ are equivalent representations), or
2. $\phi = 0$.

Proof. We analyze the kernel and image of ϕ:
1. $\ker\phi$ **is** ρ_1**-invariant**: For any $v \in \ker\phi$ and $g \in G$,

$$\phi(\rho_1(g)v) = \rho_2(g)(\phi(v)) = \rho_2(g)(0) = 0 \implies \rho_1(g)v \in \ker\phi.$$

By irreducibility of ρ_1, $\ker\phi$ is either $\{0\}$ or V_1.
2. $\operatorname{im}\phi$ **is** ρ_2**-invariant**: For any $w = \phi(v) \in \operatorname{im}\phi$,

$$\rho_2(g)w = \rho_2(g)\phi(v) = \phi(\rho_1(g)v) \in \operatorname{im}\phi.$$

By irreducibility of ρ_2, $\operatorname{im}\phi$ is either $\{0\}$ or V_2.
3. **Case analysis**:

- If $\ker\phi = V_1$, then $\phi = 0$.
- If $\ker\phi = \{0\}$, then ϕ is injective, and $\operatorname{im}\phi = V_2$ (since $\phi \neq 0$). Thus, ϕ is an isomorphism.

$\square$

Corollary 2.2.5

If $\mathbb{K}$ is algebraically closed (e.g., $\mathbb{C}$) and $V_1 = V_2$, then any intertwiner ϕ is a scalar multiple of the identity:

$$\phi = \lambda \cdot \mathrm{Id}_V.$$

Proof. Let λ be an eigenvalue of ϕ (guaranteed since $\mathbb{K}$ is algebraically closed). The map $\phi - \lambda\mathrm{Id}$ is also an intertwiner with nonzero kernel. By Schur's Lemma, $\phi - \lambda\mathrm{Id} = 0$. $\square$

1. *The lemma fails if either representation is reducible (e.g., take ϕ to be a projection).*
2. *For $\mathbb{K} = \mathbb{R}$, the corollary may not hold (e.g., irreducible representations of $\mathrm{SO}(2)$ have non-scalar intertwiners)*

3. *This is fundamental to:*
   - *Character orthogonality*
   - *Classification of irreducibles*
   - *Decomposition of tensor products*

Remark.

Let G be a group and $\rho\colon G \to \mathrm{GL}(V)$ be a finite-dimensional representation. For any basis $\mathcal{B} = \{v_i\}_{i=1}^{\dim V}$ of V, we denote by $\rho(g)_{ij}$ the matrix coefficients of $\rho(g)$ in $\mathcal{B}$, defined by:

$$\rho(g)v_j = \sum_{i=1}^{\dim V} \rho(g)_{ij} v_i, \quad \forall g \in G.$$

When G is **compact** and ρ is **unitary** (with respect to a G-invariant Hermitian inner product $\langle \cdot, \cdot \rangle$), and $\mathcal{B}$ is chosen to be an orthonormal basis, these coefficients satisfy:

1. **Unitarity condition**:
$$\rho(g^{-1})_{ij} = \overline{\rho(g)_{ji}}, \quad \forall g \in G.$$

2. **Inner product expression**:
$$\rho(g)_{ij} = \langle \rho(g)v_j, v_i \rangle.$$

3. **Orthogonality relations** (when ρ is irreducible):
$$\int_G \rho(g)_{ij} \overline{\rho'(g)_{kl}} \, dg = \frac{\delta_{\rho\rho'} \delta_{ik} \delta_{jl}}{\dim V}, \quad \text{for Haar measure } dg.$$

Theorem 2.2.6: Schur's orthogonality theorem

Let G be a compact group with normalized Haar measure dg, and let:

- (V, ρ_V) and (W, ρ_W) be finite-dimensional irreducible unitary representations of G over $\mathbb{C}$
- $\{v_i\}_{i=1}^{\dim V}$ and $\{w_k\}_{k=1}^{\dim W}$ be orthonormal bases for V and W respectively
- $\rho_V(g)_{ij} = \langle \rho_V(g)v_j, v_i \rangle_V$ and $\rho_W(g)_{kl} = \langle \rho_W(g)w_l, w_k \rangle_W$ be the matrix coefficients

Then the following orthogonality relations hold:

$$\int_G \rho_V(g)_{ij} \overline{\rho_W(g)_{kl}} \, dg = \begin{cases} \dfrac{\delta_{ik}\delta_{jl}}{\dim V} & \text{if } \rho_V \simeq \rho_W \\ 0 & \text{if } \rho_V \not\simeq \rho_W \end{cases}$$

Proof. Let G be a compact group with normalized Haar measure dg, and let (V, ρ_V) and (W, ρ_W) be finite-dimensional irreducible unitary representations.

Part 1: Construction of the Intertwiner

For any linear map $f \in \mathrm{Hom}(W, V)$, define the averaged map:

$$F := \int_G \rho_V(g) \circ f \circ \rho_W(g^{-1})\, dg$$

This map is a G-intertwiner since for any $h \in G$:

$$\begin{aligned} \rho_V(h) \circ F \circ \rho_W(h^{-1}) &= \int_G \rho_V(h)\rho_V(g) f \rho_W(g^{-1})\rho_W(h^{-1})\, dg \\ &= \int_G \rho_V(hg) f \rho_W((hg)^{-1})\, dg \\ &= F \quad \text{(by Haar measure invariance)} \end{aligned}$$

Part 2: Case of Non-isomorphic Representations
If $\rho_V \not\simeq \rho_W$, Schur's Lemma implies $F = 0$. Take $f = f_{kl}$ where:

$$f_{kl}(\omega) := \langle \omega, w_k \rangle_W v_l$$

for fixed $k \in \{1, \ldots, \dim W\}$, $l \in \{1, \ldots, \dim V\}$. Then:

$$\begin{aligned} 0 &= \langle F(w_i), v_j \rangle_V \\ &= \int_G \langle \rho_V(g) f_{kl} \rho_W(g^{-1}) w_i, v_j \rangle_V\, dg \\ &= \int_G \langle \rho_V(g) \langle \rho_W(g^{-1}) w_i, w_k \rangle_W v_l, v_j \rangle_V\, dg \\ &= \int_G \rho_V(g)_{lj} \overline{\rho_W(g)_{ki}}\, dg \end{aligned}$$

Part 3: Case of Isomorphic Representations
When $V = W$ and $\rho_V = \rho_W$, Schur's Lemma gives $F = \lambda \mathrm{Id}_V$. Taking traces:

$$\mathrm{tr}(F) = \int_G \mathrm{tr}(\rho_V(g) f \rho_V(g^{-1}))\, dg = \mathrm{tr}(f)$$

Thus $\lambda = \frac{\mathrm{tr}(f)}{\dim V}$. For $f = f_{ij}$:

$$\int_G \rho_V(g)_{lj} \overline{\rho_V(g)_{ki}}\, dg = \frac{\delta_{il}\delta_{jk}}{\dim V}$$

□

1. The proof crucially uses:

- *Unitarity to relate $\rho(g^{-1})$ and $\rho(g)^*$*
- *Haar measure invariance for the averaging trick*
- *Irreducibility via Schur's Lemma*

2. The choice of f_{kl} is designed to extract specific matrix coefficients.
3. For characters ($i = j$, $k = l$), this yields:

$$\int_G \chi_V(g) \overline{\chi_W(g)}\, dg = \delta_{[V],[W]}$$

Definition 2.2.7: L^2-Space and Orthogonal Matrix Coefficients

Let G be a compact group with normalized Haar measure dg.

- The space of **square-integrable functions** is:

$$L^2(G) := \left\{ f\colon G \to \mathbb{C} \;\middle|\; \int_G |f(g)|^2 \, dg < \infty \right\},$$

  equipped with the Hermitian inner product:

$$\langle F, G \rangle_{L^2} := \int_G \overline{F(g)} G(g) \, dg.$$

- Let $\widehat{G} = \{\rho_\alpha \colon G \to \mathrm{GL}(V_\alpha)\}_{\alpha \in A}$ be a complete set of pairwise inequivalent irreducible unitary representations.
- For each $\alpha \in A$, fix an orthonormal basis $\{v_i^{(\alpha)}\}_{i=1}^{\dim V_\alpha}$ of V_α with respect to a G-invariant inner product $\langle \cdot, \cdot \rangle_{V_\alpha}$.

The **Schur orthonormal system** consists of the normalized matrix coefficient functions:

$$\left\{ \sqrt{\dim V_\alpha} \cdot \rho_\alpha(\cdot)_{ij} \;\middle|\; \alpha \in A,\ 1 \le i, j \le \dim V_\alpha \right\},$$

where $\rho_\alpha(g)_{ij} = \langle \rho_\alpha(g) v_j^{(\alpha)}, v_i^{(\alpha)} \rangle_{V_\alpha}$.
By Schur's Orthogonality Relations, these functions form an orthonormal set in $L^2(G)$:

$$\left\langle \sqrt{\dim V_\beta} \rho_\beta(\cdot)_{kl}, \sqrt{\dim V_\alpha} \rho_\alpha(\cdot)_{ij} \right\rangle_{L^2} = \delta_{\alpha\beta} \delta_{ik} \delta_{jl}.$$

Theorem 2.2.8: Peter-Weyl

Let G be a compact group with normalized Haar measure dg, and let $\widehat{G} = \{\rho_\alpha : G \to \mathrm{U}(V_\alpha)\}_{\alpha \in A}$ be a complete set of pairwise inequivalent irreducible unitary representations. For each $\alpha \in A$, fix an orthonormal basis $\{v_j^{(\alpha)}\}_{j=1}^{d_\alpha}$ of V_α (where $d_\alpha = \dim V_\alpha$).
The normalized matrix coefficient functions:

$$\mathcal{B} = \left\{ \sqrt{d_\alpha}\rho_\alpha(\cdot)_{ij} \;\middle|\; \alpha \in A,\ 1 \le i, j \le d_\alpha \right\},$$

where $\rho_\alpha(g)_{ij} = \langle \rho_\alpha(g) v_j^{(\alpha)}, v_i^{(\alpha)} \rangle_{V_\alpha}$, form a complete orthonormal basis for the Hilbert space $L^2(G)$. That is:

1. **Orthonormality**:

$$\left\langle \sqrt{d_\alpha}\rho_\alpha(\cdot)_{ij}, \sqrt{d_\beta}\rho_\beta(\cdot)_{kl} \right\rangle_{L^2(G)} = \delta_{\alpha\beta}\delta_{ik}\delta_{jl}.$$

2. **Completeness**: Any $f \in L^2(G)$ can be expressed as

$$f(g) = \sum_{\alpha \in A} d_\alpha \sum_{i,j=1}^{d_\alpha} \langle f, \rho_\alpha(\cdot)_{ij} \rangle_{L^2(G)} \rho_\alpha(g)_{ij},$$

with convergence in the L^2-norm.

Proof. Admitted ☐

2.3 Character theory

Definition 2.3.1: Character of a Representation

Let G be a compact group and (ρ, V) a finite-dimensional continuous representation of G over $\mathbb{C}$.

- The **character** of ρ is the function $\chi_\rho \colon G \to \mathbb{C}$ defined by:

$$\chi_\rho(g) := \mathrm{Tr}_V(\rho(g)) = \sum_{i=1}^{\dim V} \langle \rho(g)v_i, v_i \rangle_V$$

  where $\{v_i\}_{i=1}^{\dim V}$ is any orthonormal basis for V with respect to a G-invariant inner product $\langle \cdot, \cdot \rangle_V$.

- When ρ is unitary (always possible for compact groups), the trace simplifies to:

$$\chi_\rho(g) = \sum_{i=1}^{\dim V} \rho(g)_{ii}$$

  where $\rho(g)_{ij}$ are the matrix coefficients in an orthonormal basis.

Proposition 2.3.2

Let G be a compact group and ρ, ρ_1, ρ_2 be finite-dimensional continuous representations over $\mathbb{C}$. Then:

1. **Class Function Property**: For all $g, h \in G$,

$$\chi_\rho(hgh^{-1}) = \mathrm{Tr}(\rho(h)\rho(g)\rho(h)^{-1}) = \chi_\rho(g).$$

   Moreover, for the contragredient representation ρ^*:

$$\chi_{\rho^*}(g) = \overline{\chi_\rho(g)}.$$

2. **Additive and Multiplicative Properties**: For representations $\rho_1 \colon G \to \mathrm{GL}(V_1)$ and $\rho_2 \colon G \to \mathrm{GL}(V_2)$,

$$\begin{aligned} \chi_{\rho_1 \oplus \rho_2} &= \chi_{\rho_1} + \chi_{\rho_2}, \\ \chi_{\rho_1 \otimes \rho_2} &= \chi_{\rho_1} \cdot \chi_{\rho_2}. \end{aligned}$$

3. **Dimension Formula**: At the identity element $e \in G$,

$$\chi_\rho(e) = \mathrm{Tr}(\mathrm{Id}_V) = \dim V \in \mathbb{N}.$$

Proof.

1. The trace is cyclic: $\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA)$. For the contragredient:

$$\chi_{\rho^*}(g) = \mathrm{Tr}(\rho(g^{-1})^\top) = \mathrm{Tr}(\rho(g^{-1})) = \overline{\chi_\rho(g)}.$$

2. For direct sums and tensor products:

   - The trace of a block matrix is the sum of traces of diagonal blocks
   - $\mathrm{Tr}(A \otimes B) = \mathrm{Tr}(A)\,\mathrm{Tr}(B)$

3. Immediate from $\rho(e) = \mathrm{Id}_V$.

$\square$

Remark.

The map

$$\chi : \text{Rep(G)} \to \text{Class functions}$$
$$\rho \mapsto \chi_\rho$$

defines a ring homomorphism The **character map**

$$\chi \colon \mathrm{Rep}(G) \to \mathcal{C}(G)$$
$$\rho \mapsto \chi_\rho$$

is a ring homomorphism when:

- $\mathrm{Rep}(G)$ is the representation ring with operations:
  - Addition: $\rho_1 + \rho_2 := \rho_1 \oplus \rho_2$
  - Multiplication: $\rho_1 \cdot \rho_2 := \rho_1 \otimes \rho_2$
- $\mathcal{C}(G)$ is the ring of complex-valued class functions with:
  - Pointwise addition: $(f_1 + f_2)(g) = f_1(g) + f_2(g)$
  - Pointwise multiplication: $(f_1 \cdot f_2)(g) = f_1(g)f_2(g)$

This follows from the character properties:

$$\chi_{\rho_1 \oplus \rho_2} = \chi_{\rho_1} + \chi_{\rho_2}$$
$$\chi_{\rho_1 \otimes \rho_2} = \chi_{\rho_1} \cdot \chi_{\rho_2}$$

Moreover, for compact G, the character map is *injective* when restricted to isomorphism classes of representations.

Theorem 2.3.3: Schur's orthonormality for character

Let G be a compact group with normalized Haar measure dg, and let (ρ_1, V_1) and (ρ_2, V_2) be finite-dimensional irreducible unitary representations of G. Their characters satisfy:

$$\langle \chi_{\rho_1}, \chi_{\rho_2} \rangle_{L^2(G)} := \int_G \overline{\chi_{\rho_1}(g)} \chi_{\rho_2}(g)\, dg = \delta_{[\rho_1],[\rho_2]},$$

where:

- $\delta_{[\rho_1],[\rho_2]} = 1$ if $\rho_1 \simeq \rho_2$ (are isomorphic), and 0 otherwise
- The inner product uses the complex conjugate of the first argument

Proof. 1. **Non-isomorphic case ($\rho_1 \not\simeq \rho_2$):** Direct consequence of Schur's Lemma applied to matrix coefficients:

$$\int_G \overline{\rho_1(g)_{ii}} \rho_2(g)_{jj}\, dg = 0 \quad \forall i, j.$$

2. **Isomorphic case ($\rho_1 \simeq \rho_2$):** Assume $\rho_1 = \rho_2$. Using orthonormality of matrix coefficients:

$$\int_G \overline{\rho_1(g)_{ii}} \rho_1(g)_{jj}\, dg = \frac{\delta_{ij}}{\dim V_1}.$$

Thus,

$$\int_G |\chi_{\rho_1}(g)|^2\, dg = \sum_{i,j=1}^{\dim V_1} \int_G \overline{\rho_1(g)_{ii}} \rho_1(g)_{jj}\, dg = \sum_{i=1}^{\dim V_1} \frac{1}{\dim V_1} = 1.$$

□

Definition 2.3.4: Space of Class functions

Let G be a compact group with normalized Haar measure dg.

- The **space of square-integrable class functions** is:

$$L^2(G)^G := \left\{ f \in L^2(G) \mid \forall h, g \in G,\ f(ghg^{-1}) = f(h) \right\},$$

  equipped with the inner product:

$$\langle f_1, f_2 \rangle := \int_G \overline{f_1(g)} f_2(g)\, dg.$$

- The **irreducible characters** are the set:

$$\left\{ \chi_\rho \mid \rho \in \widehat{G} \right\},$$

  where $\widehat{G}$ denotes the unitary dual of G (equivalence classes of irreducible unitary representations).

Theorem 2.3.5: Irreducible Characters as Orthonormal Basis

Let G be a compact group with normalized Haar measure dg, and let $\widehat{G}$ denote the set of equivalence classes of irreducible unitary representations of G.
The set of irreducible characters

$$\left\{\chi_\rho \mid \rho \in \widehat{G}\right\}$$

forms a complete orthonormal basis for the Hilbert space $L^2(G)^G$ of square-integrable class functions, where:

1. **Orthonormality**: For $\rho, \rho' \in \widehat{G}$,

$$\langle \chi_\rho, \chi_{\rho'} \rangle_{L^2(G)} = \int_G \overline{\chi_\rho(g)} \chi_{\rho'}(g)\, dg = \delta_{[\rho],[\rho']}$$

2. **Completeness**: Every class function $f \in L^2(G)^G$ has a convergent expansion

$$f = \sum_{\rho \in \widehat{G}} \langle f, \chi_\rho \rangle \chi_\rho$$

with convergence in the L^2-norm.

Proof. Let $f \in L^2(G)^G$ be a square-integrable class function on a compact group G.

1. **Peter-Weyl Expansion**: By the Peter-Weyl theorem, f admits an expansion in matrix coefficients:

$$f(h) = \sum_{\rho \in \widehat{G}} \sum_{i,j=1}^{\dim \rho} c_{\rho,ij} \rho(h)_{ij}, \quad \text{where } \rho(h)_{ij} = \langle \rho(h) v_j, v_i \rangle.$$

2. **Class Function Property**: Since f is a class function, for all $g, h \in G$:

$$f(h) = \int_G f(ghg^{-1})\, dg = \sum_{\rho,i,j} c_{\rho,ij} \int_G \rho(ghg^{-1})_{ij}\, dg.$$

3. **Intertwining Argument**: For each ρ, define the operator:

$$F_\rho := \int_G \rho(g)\rho(h)\rho(g^{-1})\, dg.$$

This satisfies the intertwining property:

$$\rho(k) F_\rho = F_\rho \rho(k) \quad \forall k \in G.$$

By Schur's Lemma, since ρ is irreducible:

$$F_\rho = \lambda(h) \mathrm{Id}_V.$$

4. **Trace Calculation**: Taking traces on both sides:

$$\begin{aligned}\operatorname{Tr}(F_\rho) &= \int_G \operatorname{Tr}(\rho(g)\rho(h)\rho(g^{-1}))\,dg \\ &= \int_G \chi_\rho(h)\,dg = \chi_\rho(h) \\ &= \lambda(h)\dim V.\end{aligned}$$

   Thus $\lambda(h) = \dfrac{\chi_\rho(h)}{\dim V}$.

5. **Final Expansion**: Substituting back:

$$f(h) = \sum_{\rho\in\widehat{G}} \left\langle f, \frac{\chi_\rho}{\dim V}\right\rangle \chi_\rho(h),$$

   where the coefficients are given by the L^2 inner product.

□

Remark.

- If $|G| < \infty$, then clearly $\dim L^2(G)^G < \infty$ because:
  - The space $L^2(G)^G$ of square-integrable class functions has dimension equal to the number of conjugacy classes
  - Finite groups have finitely many conjugacy classes

  Hence, $\{\chi_\rho : \rho \in \widehat{G}\}$ is a finite set, i.e., G has **finitely many** irreducible representations.
- If G is infinite, then:
  - $L^2(G)^G$ is typically infinite-dimensional
  - This implies G must have infinitely many irreducible unitary representations
    - For compact groups: By Peter-Weyl theorem
    - For non-compact groups: Via Plancherel theorem (continuous decomposition)

Corollary 2.3.6

Let G be a compact group. For any finite-dimensional representation $\rho\colon G \to \mathrm{GL}(V)$ of G on V, we have a decomposition:

$$\rho \sim \bigoplus_{\alpha\in A} \rho_\alpha^{\oplus m_\alpha}$$

where:

- $\{\rho_\alpha\}_{\alpha\in A}$ are irreducible representations of G
- m_α are the corresponding multiplicities

We denote by $[\rho, \rho_\alpha] = m_\alpha$ the multiplicity of the irreducible representation ρ_α in the decomposition of ρ. For all $\alpha \in A$, these multiplicities can be computed via the inner product of characters:

$$[\rho, \rho_\alpha] = (\chi_\rho, \chi_{\rho_\alpha})$$

Proof. By definition of the character of ρ, we have:

$$\chi_\rho = \sum_{\beta\in A} m_\beta \chi_{\rho_\beta}$$

Applying the inner product with χ_{ρ_α} and using Schur's orthogonality relations for characters:

$$(\chi_\rho, \chi_{\rho_\alpha}) = \left(\sum_{\beta\in A} m_\beta \chi_{\rho_\beta}, \chi_{\rho_\alpha}\right) = \sum_{\beta\in A} m_\beta (\chi_{\rho_\beta}, \chi_{\rho_\alpha}) = m_\alpha$$

where the last equality follows since $(\chi_{\rho_\beta}, \chi_{\rho_\alpha}) = \delta_{\beta\alpha}$ by Schur's orthogonality. □

Remark.

The structure of the representation ring $(\mathrm{Rep}(G), \oplus, \otimes)$ is completely determined for a compact group G, since we know:

- The complete list of irreducible representations $\{\rho_\alpha\}_{\alpha\in A}$
- The decomposition of their tensor products into irreducibles:

  $$\rho_\alpha \otimes \rho_\beta \cong \bigoplus_{\gamma\in A} N_{\alpha\beta}^\gamma \rho_\gamma$$

  where $N_{\alpha\beta}^\gamma$ are the multiplicity coefficients

Thanks to character theory, this information reduces to knowing **the list of irreducible characters** $\{\chi_{\rho_\alpha}\}_{\alpha\in A}$. Indeed, the multiplicities can be computed via:

$$[\rho_\alpha \otimes \rho_\beta; \rho_\gamma] = (\chi_{\rho_\alpha\otimes\rho_\beta}, \chi_{\rho_\gamma}) = (\chi_{\rho_\alpha}\chi_{\rho_\beta}, \chi_{\rho_\gamma})$$

where $(\cdot, \cdot)$ denotes the inner product on class functions.

2.3.1 Finite Group

Let G be a finite group with $|G| < \infty$. Then:

- The space of class functions $L^2(G)^G$ is finite-dimensional:

$$\dim L^2(G)^G < \infty$$

- Moreover, we have the following equalities:

$$\dim L^2(G)^G = (\widehat{G}) = \text{number of inequivalent irreducible representations of } G$$

$$= \text{number of conjugacy classes of } G$$

The character table of a finite group G displays the irreducible characters evaluated on conjugacy classes:

G	$[g_1]$	$[g_2]$	$\cdots$	$[g_k]$
χ_1	$\chi_1(g_1)$	$\chi_1(g_2)$	$\cdots$	$\chi_1(g_k)$
χ_2	$\chi_2(g_1)$	$\chi_2(g_2)$	$\cdots$	$\chi_2(g_k)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
χ_r	$\chi_r(g_1)$	$\chi_r(g_2)$	$\cdots$	$\chi_r(g_k)$

Table 2.1: Character table of G, where $[g_i]$ are conjugacy class representatives and χ_j are irreducible characters. For finite groups, $r = k$ equals the number of conjugacy classes.

2.4 Symmtries in Quantum Physics and projective representations

In quantum physics, the states of a system are described by elements of the *projective space*

$$P(V) = V/\sim,$$

where V is a complex Hilbert space and $\sim$ is the equivalence relation defined for nonzero vectors:

$$v \sim w \quad \text{if and only if} \quad v = \lambda w \;\text{ for some }\; \lambda \in \mathbb{C}^\times.$$

Here, $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ denotes the multiplicative group of nonzero complex numbers.

Alternatively, states can be viewed as *rays* in V, where two vectors represent the same physical state if they are scalar multiples of each other.

In the study of group actions on quantum states, one should consider group homomorphisms of the form

$$G \to \big(V\big),$$

where V is the projective space associated with the Hilbert space V, rather than homomorphisms $G \to \mathrm{GL}(V)$.

There exists a natural surjective group homomorphism

$$\mathrm{GL}(V) \xrightarrow{f} (V),$$

defined for all $A \in \mathrm{GL}(V)$ and all nonzero $v \in V \setminus \{0\}$ by

$$f(A)\big([v]\big) = [A(v)].$$

Here, $[v] \in V$ denotes the equivalence class of v under the projective equivalence relation.

The kernel of f consists of scalar multiples of the identity:

$$\ker(f) = \big\{A \in \mathrm{GL}(V) \mid f(A) = \mathrm{id}_{\mathbb{P}(V)}\big\} = \mathbb{C}^\times \cdot \mathrm{id}_V,$$

where $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ and id_V is the identity map on V.

By the 1st isomorphism theorem:

- $\ker(f) \cong \mathbb{C}^\times \trianglelefteq \mathrm{GL}(V)$
- $\mathrm{Aut}(\mathbb{P}(V)) \cong \mathrm{GL}(V)/\mathbb{C}^\times = \mathrm{PGL}(V)$

Definition 2.4.1: Projective representation

Let G be a topological group and V a finite-dimensional $\mathbb{C}$-vector space. A **projective representation** of G on V is a continuous group homomorphism:

$$\rho\colon G \to \mathrm{PGL}(V),$$

where $\mathrm{PGL}(V) := \mathrm{GL}(V)/\mathbb{C}^\times$ is the projective linear group, and the quotient is taken with respect to the subspace topology induced by the natural projection $\pi\colon \mathrm{GL}(V) \to \mathrm{PGL}(V)$.

Definition 2.4.2

Let V be a finite-dimensional $\mathbb{C}$-vector space. The **projective linear group** is defined by the quotient:

$$\pi\colon \mathrm{GL}(V) \to \mathrm{PGL}(V) := \mathrm{GL}(V)/\mathbb{C}^\times$$

where:

- $\mathbb{C}^\times$ acts on $\mathrm{GL}(V)$ by scalar multiplication: $\lambda \cdot A = \lambda A$ for $\lambda \in \mathbb{C}^\times$, $A \in \mathrm{GL}(V)$
- π is the canonical projection map, continuous in the quotient topology
- For $V = \mathbb{C}^n$, we write $\mathrm{PGL}(n, \mathbb{C})$

Proposition 2.4.3

Let G be a topological group and (ρ, V) a finite-dimensional continuous linear representation of G, where:

$$\rho\colon G \to \mathrm{GL}(V)$$

is a continuous homomorphism. Then the composition

$$\pi \circ \rho\colon G \to \mathrm{PGL}(V)$$

defines a continuous projective representation of G, where:

- $\pi\colon \mathrm{GL}(V) \to \mathrm{PGL}(V) = \mathrm{GL}(V)/\mathbb{C}^\times$ is the canonical projection
- The action is given by $g \cdot [v] = [\rho(g)v]$ for $g \in G$, $[v] \in \mathbb{P}(V)$

Proof. The composition $\pi \circ \rho\colon G \to \mathrm{PGL}(V)$ is a continuous projective representation because:

1. **Group Homomorphism Property**:
   - $\rho\colon G \to \mathrm{GL}(V)$ is a group homomorphism by assumption
   - $\pi\colon \mathrm{GL}(V) \to \mathrm{PGL}(V)$ is a group homomorphism as it is the canonical projection
   - Therefore, their composition preserves the group operation:

   $$(\pi \circ \rho)(gh) = \pi(\rho(gh)) = \pi(\rho(g)\rho(h)) = \pi(\rho(g))\pi(\rho(h)) = (\pi \circ \rho)(g)(\pi \circ \rho)(h)$$

2. **Continuity**:
   - ρ is continuous by definition of a linear representation
   - π is continuous in the quotient topology
   - The composition of continuous maps is continuous

3. **Projective Action**: The induced action on $\mathbb{P}(V)$ is well-defined since scalar multiples act trivially:

   $$[\rho(g)(\lambda v)] = [\lambda \rho(g)v] = [\rho(g)v] \quad \forall \lambda \in \mathbb{C}^\times, v \in V$$

□

Question: Given a projective representation $\rho : G \to \mathrm{PGL}(V)$, is there a lift of ρ to a linear representation $\hat{\rho} : G \to \mathrm{GL}(V)$? That is, does there exist a linear representation $\hat{\rho} : G \to \mathrm{GL}(V)$ such that:

$$\rho = \pi \circ \hat{\rho}$$

Let $\rho : G \to \mathrm{PGL}(V)$ be a projective representation. For each $g \in G$, pick a representative $r(g) \in \mathrm{GL}(V)$ of $\rho(g) \in \mathrm{PGL}(V)$, i.e., an element $r(g) \in \mathrm{GL}(V)$ such that:

$$\pi(r(g)) = \rho(g)$$

Question becomes: Can $r : G \to \mathrm{GL}(V)$, $g \mapsto r(g)$ be made into a group homomorphism?

Indeed, $\forall g, g' \in G$

$$\begin{aligned}\pi(r(g)r(g')) &= \rho(g)\rho(g') = \rho(gg')\\ \pi(r(gg')) &= \rho(gg')\end{aligned}$$

Hence $\forall g, g' \in G$, $\pi(r(g)r(g')) = \pi(r(gg'))$. It follows that there exists a map $\Omega : G \times G \to \mathbb{C}^\times$ such that

$$\forall g, g' \in G : r(g)r(g') = \Omega(g, g')r(gg')$$

Question becomes: Can we choose the lift $r\colon G \to \mathrm{GL}(V)$ in such a way that the associated cocycle $\Omega(g, g') \equiv 1$ for all $g, g' \in G$?

By the associativity of the group law in G, we have for all $g, g', g'' \in G$:

$$\begin{aligned}r(g)r(g')r(g'') &= \Omega(g', g'')r(g)r(g'g'')\\ &= \Omega(g', g'')\Omega(g, g'g'')r(gg'g'')\\ &= \Omega(g, g')r(gg')r(g'')\\ &= \Omega(g, g')\Omega(gg', g'')r(gg'g'')\end{aligned}$$

Comparing both expressions for $r(g)r(g')r(g'')$, we obtain the **2-cocycle condition**:

$$\Omega(g, g')\Omega(gg', g'') = \Omega(g', g'')\Omega(g, g'g'')$$

Definition 2.4.4: Group of 2-cocylce

Let G be a group and $\mathbb{C}^\times$ the multiplicative group of nonzero complex numbers. The **group of 2-cocycles** is defined as:

$$Z^2(G, \mathbb{C}^\times) := \left\{ \Omega\colon G \times G \to \mathbb{C}^\times \;\middle|\; \begin{array}{r} \forall g, g', g'' \in G : \\ \Omega(g, g')\Omega(gg', g'') = \Omega(g', g'')\Omega(g, g'g'') \end{array} \right\}$$

This forms an abelian group under pointwise multiplication:

$$(\Omega_1 \cdot \Omega_2)(g, g') := \Omega_1(g, g') \cdot \Omega_2(g, g')$$

with identity element the constant function 1 and inverses $\Omega^{-1}(g, g') = \Omega(g, g')^{-1}$.

Proposition 2.4.5

Let $\rho\colon G \to \mathrm{PGL}(V)$ be a projective representation with two lifts to $\mathrm{GL}(V)$:

- Original lift: $r(g)r(g') = \Omega(g, g')r(gg')$ with $\Omega \in Z^2(G, \mathbb{C}^\times)$
- New lift: $r'(g) = \alpha(g)r(g)$ for some $\alpha\colon G \to \mathbb{C}^\times$

Then the new cocycle Ω' satisfies:

$$\Omega'(g, g') = \frac{\alpha(g)\alpha(g')}{\alpha(gg')}\Omega(g, g')$$

Proof. For any $g, g' \in G$, we compute:

$$\begin{aligned}
r'(g)r'(g') &= \alpha(g)r(g) \cdot \alpha(g')r(g') \\
&= \alpha(g)\alpha(g') \cdot r(g)r(g') \\
&= \alpha(g)\alpha(g')\Omega(g, g')r(gg') \\
&= \frac{\alpha(g)\alpha(g')}{\alpha(gg')}\Omega(g, g') \cdot \alpha(gg')r(gg') \\
&= \underbrace{\left(\frac{\alpha(g)\alpha(g')}{\alpha(gg')}\Omega(g, g')\right)}_{\Omega'(g,g')} r'(gg')
\end{aligned}$$

□

Question becomes: Can we find a map $\alpha\colon G \to \mathbb{C}^\times$ such that $\Omega' = 1$? That is, which 2-cocycles $\Omega \in Z^2(G, \mathbb{C}^\times)$ can be expressed in the form:

$$\Omega(g, g') = \frac{\alpha(g)\alpha(g')}{\alpha(gg')} \quad \forall g, g' \in G$$

Definition 2.4.6: Coboundary Map

For any map $\alpha\colon G \to \mathbb{C}^\times$, define its **coboundary** as:

$$\partial\alpha\colon G \times G \to \mathbb{C}^\times, \quad (g, g') \mapsto \frac{\alpha(g)\alpha(g')}{\alpha(gg')}$$

Proposition 2.4.7

Every coboundary $\partial\alpha$ satisfies the 2-cocycle condition:

$$\partial\alpha \in Z^2(G, \mathbb{C}^\times)$$

Proof. Direct verification:

$$\begin{aligned}\partial\alpha(g,g')\partial\alpha(gg',g'') &= \frac{\alpha(g)\alpha(g')}{\alpha(gg')}\cdot\frac{\alpha(gg')\alpha(g'')}{\alpha(gg'g'')}\\ &= \frac{\alpha(g')\alpha(g'')}{\alpha(g'g'')}\cdot\frac{\alpha(g)\alpha(g'g'')}{\alpha(gg'g'')}\\ &= \partial\alpha(g',g'')\partial\alpha(g,g'g'')\end{aligned}$$

□

Proposition 2.4.8

Given $\Omega \in Z^2(G,\mathbb{C}^\times)$, the following are equivalent:

1. Ω is a coboundary: $\exists\alpha\colon G \to \mathbb{C}^\times$ with $\Omega = \partial\alpha$
2. The projective representation ρ lifts to a linear representation $\tilde{\rho}\colon G \to \mathrm{GL}(V)$

When these hold, the adjusted lift $\tilde{\rho}(g) := \alpha(g)^{-1}r(g)$ becomes a genuine homomorphism.

Proposition 2.4.9

The coboundary map ∂ is a group homomorphism

$$\partial\colon C^1(G,\mathbb{C}^\times) \to Z^2(G,\mathbb{C}^\times)$$

where $C^1(G,\mathbb{C}^\times) = \{\alpha\colon G \to \mathbb{C}^\times\}$ is the group of 1-cochains. Its image

$$B^2(G,\mathbb{C}^\times) := \mathrm{im}(\partial) = \{\partial\alpha \mid \alpha\colon G \to \mathbb{C}^\times\} \trianglelefteq Z^2(G,\mathbb{C}^\times)$$

forms an abelian subgroup of 2-coboundaries, normal in $Z^2(G,\mathbb{C}^\times)$ under pointwise multiplication.

Definition 2.4.10

The **second cohomology group** of G with coefficients in $\mathbb{C}^\times$ is:

$$H^2(G,\mathbb{C}^\times) := Z^2(G,\mathbb{C}^\times)/B^2(G,\mathbb{C}^\times)$$

where:

- $Z^2(G,\mathbb{C}^\times)$ = group of 2-cocycles
- $B^2(G,\mathbb{C}^\times)$ = group of 2-coboundaries

Question becomes: Is $H^2(G,\mathbb{C}^\times) = \{1\}$?
If $H^2(G,\mathbb{C}^\times) = \{1\}$, then any 2-cocycle $\Omega \in Z^2(G,\mathbb{C}^\times)$ is of the form

$$\Omega(g,g') = \frac{\alpha(gg')}{\alpha(g)\alpha(g')}$$

and it suffices to take $r'(g) = \alpha(g)r(g)$ to get $\Omega'(g, g') = 1$.

If $H^2(G, \mathbb{C}^\times) \neq \{1\}$, then G admits projective representations that cannot be lifted to linear representations of G.

Proposition 2.4.11

Let G be a topological group with $H^2(G, \mathbb{C}^\times) \neq \{1\}$.
Let $\rho : G \to \mathrm{PGL}(V)$ be a projective representation of G on V.
Then, there exist:

1. A group $\hat{G}$
2. A surjective group homomorphism $p : \hat{G} \to G$
3. A linear representation $\hat{\rho} : \hat{G} \to \mathrm{GL}(V)$ such that $\pi \circ \hat{\rho} = \rho \circ p$
4. $\ker(p) = \mathbb{C}^\times$ (i.e., $\hat{G}$ is a central extension of G by $\mathbb{C}^\times$)

Proof. Let G be a topological group with $H^2(G, \mathbb{C}^\times) \neq \{1\}$ and $\rho\colon G \to \mathrm{PGL}(V)$ a projective representation with associated cocycle $\Omega \in Z^2(G, \mathbb{C}^\times)$ where $[\Omega] \neq 1$.

1. **Group Construction**: Define $\hat{G} := G \times \mathbb{C}^\times$ with multiplication

$$(g_1, z_1) \cdot (g_2, z_2) := (g_1 g_2, z_1 z_2 \Omega(g_1, g_2))$$

   This operation is associative because:

$$\begin{aligned}((g_1, z_1)(g_2, z_2))(g_3, z_3) &= (g_1 g_2 g_3, z_1 z_2 z_3 \Omega(g_1, g_2)\Omega(g_1 g_2, g_3))\\(g_1, z_1)((g_2, z_2)(g_3, z_3)) &= (g_1 g_2 g_3, z_1 z_2 z_3 \Omega(g_2, g_3)\Omega(g_1, g_2 g_3))\end{aligned}$$

   which are equal by the cocycle condition $\Omega(g_1, g_2)\Omega(g_1 g_2, g_3) = \Omega(g_2, g_3)\Omega(g_1, g_2 g_3)$.

2. **Projection**: The map $p\colon \hat{G} \to G$ given by $(g, z) \mapsto g$ is a continuous surjective homomorphism since:

$$p((g_1, z_1)(g_2, z_2)) = g_1 g_2 = p(g_1, z_1)p(g_2, z_2)$$

3. **Linear Representation**: Define $\hat{\rho}\colon \hat{G} \to \mathrm{GL}(V)$ by $(g, z) \mapsto zr(g)$. This is a homomorphism because:

$$\begin{aligned}\hat{\rho}((g_1, z_1))\hat{\rho}((g_2, z_2)) &= z_1 r(g_1) z_2 r(g_2)\\&= z_1 z_2 \Omega(g_1, g_2) r(g_1 g_2)\\&= \hat{\rho}((g_1 g_2, z_1 z_2 \Omega(g_1, g_2)))\\&= \hat{\rho}((g_1, z_1)(g_2, z_2))\end{aligned}$$

4. **Commutativity**: The diagram commutes since:

$$\pi \circ \hat{\rho}(g, z) = \pi(zr(g)) = \pi(r(g)) = \rho(g) = \rho \circ p(g, z)$$

5. **Kernel**: $\ker p = \{(e,z) \mid z \in \mathbb{C}^\times\} \cong \mathbb{C}^\times$ is central in $\hat{G}$ because:

$$(g,1)(e,z) = (g,z) = (e,z)(g,1)$$

□

Theorem 2.4.12: Theorem Name

For any simply-connected Lie group G, the second cohomology group is trivial:

$$H^2(G,\mathbb{C}^\times) = \{1\}$$

Proof. Admitted □

Let G be a multiply connected Lie group with universal cover

$$p : \hat{G} \to G.$$

Thus for any projective representation $\rho : G \to \mathrm{PGL}(V)$, we have

- $\rho \circ p$ is a projective representation of $\hat{G}$.
- But since $\hat{G}$ is a simply-connected Lie group, the theorem implies that $H^2(\hat{G},\mathbb{C}^\times) = \{1\}$. Hence, $\rho \circ p$ can be lifted to a linear representation $\hat{\rho} : \hat{G} \to \mathrm{GL}(V)$.

Example.

> $\mathrm{SU}(2) \xrightarrow{p} \mathrm{SO}(3)$